CAREGIVER'S NUTRITION KNOWLEDGE, FEEDING PRACTICES AND NUTRITIONAL STATUS OF CHILDREN 6-23 MONTHS AT KOMAMBOGA HEALTH CENTRE III, KAWEMPE DIVISION, KAMPALA

BY

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A RESEARCH REPORT SUBMITTED TO THE DEPARTMENT OF NUTRITIONAL SCIENCE AND DIETETICS IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF BACHELOR OF SCIENCE IN HUMAN NUTRITION AND DIETETICS OF KYAMBOGO UNIVERSITY

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DECLARATION

I, NABIRYE FLORENCE, hereby declare that this report is my original work and it has not been submitted to any other institution of higher learning for any award.

Signature: .....

Date: .....

APPROVAL

This is to certify that the information included in this research report, entitled “Caregiver’s Nutrition Knowledge, Feeding Practices, and Nutritional Status of Children 6-23 Months at Komamboga Health Centre III, Kawempe Division, Kampala,” was conducted under my supervision. It is now ready for submission to the Academic Board of the Department of Nutritional Science and Dietetics of Kyambogo University under the School of Vocational Studies.

DR. MANGUSHO GILBERT

Signature: .....

Date: .....

DEDICATION

I dedicate this report to my husband, Mr. Kirigwajjo Moses, whose unwavering support, encouragement, and love have been my constant source of strength throughout this journey.

ACKNOWLEDGMENT

I would like to acknowledge the Almighty God for enabling me not only during this proposal and report writing process, but also for my degree program so far.

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ACRONYMS AND ABBREVIATIONS

ASFs	Animal Source Foods
DDS	Dietary Diversity Score
FAO	Food and Agriculture Organization
MAD	Minimum Acceptable Diet
MDD	Minimum Dietary Diversity
MMF	Minimum Meal Frequency
NS	Nutritional Status
SDGs	Sustainable Development Goals
UBOS	Uganda Bureau of Statistics
UDHS	Uganda Demographic Health Survey
UN	United Nations
UNICEF	United Nations Children's Fund
WHO	World Health Organization

OPERATIONAL DEFINITIONS OF TERMS

Caregiver: the primary person responsible for the day-to-day feeding and physical care of a child aged 6–23 months. This includes biological mothers (primary), fathers, or alternative guardians (such as grandmothers or housekeepers) who have spent the majority of the day with the child for at least six (Mekonen et al., 2024).

Nutrition Knowledge: This refers to the caregiver's level of understanding of recommended IYCF guidelines, including the age at which solids are introduced, meal frequency, dietary diversity, and food consistency (Gizaw & Tesfaye, 2019).

Feeding Practices: These are the actual behaviors and actions performed by the caregiver in feeding the child (Mekonen et al., 2024).

Nutritional Status: The physiological state of the child resulting from the intake and utilization of nutrients, assessed through anthropometric measurements (Cheng et al., 2023).

ABSTRACT

Background: Suboptimal infant and young child feeding practices in Uganda remain a leading direct driver of early childhood malnutrition, particularly in densely populated peri-urban informal settlements. This study evaluated the relationship between caregiver knowledge, complementary feeding practices, and the nutritional status of children aged 6-23 months at Komamboga Health Centre III, Kawempe Division, Kampala.

Methods: A health facility-based cross-sectional study was conducted among 170 caregivers and children aged 6–23 months. Quantitative data were collected using structured questionnaires. Child anthropometric measurements were obtained using calibrated equipment and converted into Z-scores based on the WHO 2006 Child Growth Standards. Data were analyzed using SPSS Version 26.0. Bivariate Chi-Square were executed to evaluate statistical associations between IYCF indicators and nutritional outcomes ($p < 0.05$).

Results: Assessment of caregiver nutritional knowledge revealed that nearly half demonstrated moderate knowledge, while a quarter possessed high knowledge, with an overall mean knowledge score of 61.4% (±14.8). Regarding complementary feeding practices, only 28.2% of children achieved a Minimum Acceptable Diet. Although 61.8% met the age-appropriate Minimum Meal Frequency, only 36.5% achieved Minimum Dietary Diversity. Diets were mainly dominated by starchy staples, whereas intake of nutrient-dense animal-source foods was markedly low. Anthropometric assessment revealed high levels of child undernutrition. Male children exhibited higher rates of growth deficits across all indices compared to females. Bivariate statistical testing demonstrated that meeting MDD was significantly associated with lower rates of stunting ($\chi^2 = 8.42$, $p = 0.004$) and underweight ($\chi^2 = 5.16$, $p = 0.023$). Similarly, MAD compliance was protective against stunting ($\chi^2 = 7.89$, $p = 0.005$). Consumption of animal-source foods was strongly associated with reduced stunting ($\chi^2 = 9.14$, $p = 0.002$). Meal frequency alone was associated with reduced acute wasting ($\chi^2 = 4.12$, $p = 0.042$) but showed no significant relationship with linear stunting ($X^2 = 1.52$, $p = 0.218$).

Conclusion and Recommendations: While caregiver nutritional knowledge is overall moderate, complementary feeding quality remains critically poor, characterized by adequate meal frequency but severe deficits in dietary diversity and animal-protein intake driven by urban food price inflation and maternal time poverty. Healthcare providers at primary health facilities must strengthen targeted maternal counseling that emphasizes dietary quality and affordable local protein enrichment (such as mukene silverfish powder and groundnut paste).

CHAPTER ONE: INTRODUCTION

1.1 Background

The World Health Organization defines complementary feeding as the process of providing foods in addition to milk when breast milk or milk formula alone are no longer adequate to meet nutritional requirements c The shift from exclusive breastfeeding to family foods, known as complementary feeding, often occurs between 6 and 23 months of age, although breastfeeding may persist until two years of age and beyond (Miniello et al., 2021). This is a critical period of growth during which nutrient deficiencies and illnesses contribute globally to higher rates of undernutrition among children under five years of age (Capra et al., 2024). If complementary foods are not introduced around the age of 6 months, or if they are given inappropriately, an infant's growth may falter (UNICEF, 2025).

Globally, only two-thirds of 6-8-month-olds receive any solid food; of these, 1 in 2 receive a minimum meal frequency, and less than 1 in 3 receive minimum dietary diversity. Considering both minimum meal frequency and minimum diet diversity, only about 1 in 6 children receives a minimally acceptable diet (Barros et al., 2021). In developing countries, 1 in 5 children aged 6 to 23 months is fed the minimum recommended diverse diet (UNICEF, 2019). According to the 2022 Uganda demographic and health survey (UDHS), 7% of children aged 6–23 months met the minimum standards concerning all three infant and young child feeding (IYCF) practices (breastfeeding status, number of food groups, and times) they fed during the day or night before the survey (Uganda Demographic And Health Survey [UDHS], 2022).

For a child, the first two years are the period during which linear faltering of growth is most prevalent and the period when the process of becoming stunted is almost complete (Geda et al., 2021). Stunting, a sub-form of malnutrition responsible for 14% of global childhood deaths, also heavily debilitates cognitive growth and physical capabilities, leaving a person's future less gainful than it would otherwise have been (Danaei et al., 2016). There are many risk factors for stunting, and studies most often corroborate complementary feeding practices as a significant risk factor (Miniello et al., 2021). These reports of inappropriate and inadequate complementary feeding practices may be responsible for the world's 21.9% of stunted children (FAO, 2020).

Inadequate dietary diversity and meal frequency lead children to undernutrition, morbidity, and mortality (Tesfa et al., 2022), impaired cognitive development, and low economic productivity in long life (Gizaw & Tesfaye, 2019). Malnutrition has been responsible, directly or indirectly, for 47% of the deaths annually among children under five in Uganda (UNICEF, 2025). Over two-thirds of these deaths, which were often associated with inappropriate feeding practices, occur during the first year of life (Cheng et al., 2023).

The WHO core indicators of optimum nutrition for children above 6 months include continued breastfeeding, the introduction of age-appropriate complementary foods, minimum dietary diversity, minimum meal frequency, and minimum acceptable diet (Marriott et al., 2012). Among the key indicators, minimum dietary diversity is measured based on consumption of at least 5 or more varieties of foods from the 8 food groups in 24 hours, Minimum meal frequency is measured in terms of receiving complementary foods the minimum recommended number of times in 24 hours (for breastfed children, the frequency should be at least 2 times for 6 to 8 months, and at least 3 times for children 9 to 23 months of age while for non-breastfed children, it should be at least 4 times) (Atim et al., 2024). Hence, it is important to assess children's diets to determine the level at which these recommendations are met.

1.2 Problem Statement

Malnutrition remains the leading underlying cause of child mortality globally, contributing to nearly 45% of all deaths in children under five (WHO, 2023). Undernutrition (stunting at 30% and wasting at 7.3%) accounts for at least half of all the deaths annually in children under five years of age in sub-Saharan Africa (Khan et al., 2019). Results from the previous Uganda Demographic Health Surveys (2022) indicated that 7% of all infants were born with low birth weight (< 2500 g), 26% were stunted, 11% were underweight, and 3.5% were wasted (Uganda Demographic And Health Survey [UDHS], 2022).

Several nutrition initiatives have been implemented in Uganda to improve the nutritional status outcomes for children at both the health facility and community levels. These initiatives include promotion of breastfeeding, nutrition education, and counselling on maternal, infant, and young child feeding, growth monitoring, and promotion of integrated management of acute malnutrition, micronutrient supplementation, promotion of consumption of fortified foods, and promotion of household diet diversity (WHO, 2023). In addition, communication and advocacy initiatives have been employed, for example, the Uganda Nutrition Advocacy and Communication Strategy to promote awareness of nutrition in Uganda.

Despite these initiatives, Infant and Young Child Feeding (IYCF) among children 6-23 months is still suboptimal in the country. About 66% of children under the age of 6 months are exclusively breastfed, and six in seven children aged 6–23 months (about 85 percent) do not consume a minimally adequate diet; that is, they do not consume food from at least three food groups at least twice a day (UBOS, 2023).

In urban settings such as Kawempe Division, Kampala, these challenges are compounded by high population density, varying maternal employment patterns, and limited access to diverse food groups (Mekonen et al., 2024). Komamboga Health Centre III, being the only public health facility in the whole of Kawempe Division, which is characterized by many slums, continues to record high cases of preventable nutritional deficiencies like iron deficiency anaemia and wasting, suggesting that standard nutritional counselling is not effectively translating into practice at the household level. Failure to address these feeding inadequacies will lead to a continued surge in childhood morbidity and early childhood mortality and directly undermine Uganda's progress towards Sustainable Development Goal (SDG) 2 (Zero hunger) and SDG 3 (Good health and well-being).

Lastly, while the immediate causes of malnutrition, namely inadequate dietary intake and repeated bouts of infectious diseases, are well-documented, there is a significant lack of localized evidence regarding the socio-behavioural drivers specific to the Kawempe community. Furthermore, it remains unclear how much maternal knowledge, deep-seated cultural myths, and economic resource constraints interact to prevent mothers from adhering to recommended feeding schedules. Without understanding these specific care-based practices, interventions remain generic and ineffective. The purpose of this study was therefore to determine caregiver complementary feeding practices and nutritional status among children 6-23 months at Komamboga Health Centre III, Kawempe Division, Kampala.

1.3 Research Objectives

1.3.1 General Objective

To assess the relationship between caregiver knowledge, complementary feeding practices, and the nutritional status of children aged 6-23 months at Komamboga Health Centre III, Kawempe Division, Kampala.

1.3.2 Specific Objectives

i. To evaluate the level of nutritional knowledge regarding complementary feeding among caregivers at Komamboga Health Centre III, Kawempe Division, Kampala.
ii. To assess the complementary feeding practices among caregivers of children aged 6-23 months at Komamboga Health Centre III, Kawempe Division, Kampala.
iii. To determine the nutritional status of children aged 6-23 months attending Komamboga Health Centre III, Kawempe Division, Kampala.
iv. To establish the association between caregiver knowledge, feeding practices, and the nutritional status of children aged 6-23 months at Komamboga Health Centre III.

1.4 Research Questions

i. What is the level of nutritional knowledge regarding complementary feeding among caregivers at Komamboga Health Centre III?
ii. What are the complementary feeding practices (diversity, frequency, and timing) among caregivers of children aged 6–23 months at Komamboga Health Centre III?
iii. What is the nutritional status (stunting, wasting, and underweight) of children aged 6–23 months attending Komamboga Health Centre III?
iv. What is the relationship between caregiver complementary feeding practices and the nutritional status of children aged 6-23 months at Komamboga Health Centre III?

1.5 Significance of the Study

The study provided facility-specific data on the nutritional status and feeding gaps among children in the Komamboga catchment area. Rather than relying on general health education, nurses and nutritionists can empowered to offer targeted counseling based on the actual household practices. Furthermore, by evaluating the relationship between specific feeding behaviors and nutritional outcomes, the study aims to uncover which localized practices are the primary drivers of growth faltering in this specific population.

The results serve as a situational analysis tool for the Kampala Capital City Authority (KCCA) health department. It provided practical evidence to justify the allocation of resources for nutrition-specific interventions, such as vitamin A supplementation drives or community-based nutrition workshops in Komamboga.

As Uganda strives to meet Sustainable Development Goal (SDG) 2, which is ending all forms of malnutrition by 2030, this study provided a localized snapshot of how urban pressures in Kampala affect national nutrition targets. It also influences the refinement of the Uganda Nutrition Action Plan (UNAP) regarding urban-specific feeding challenges.

1.6 Conceptual Framework

The conceptual framework for this study is based on the premise that a child's nutritional status is the result of a multi-dimensional causal chain. It posits that Caregiver Nutrition Knowledge serves as a foundational determinant that shapes Complementary Feeding Practices, such as dietary diversity, meal frequency, and the timely introduction of solid foods.

In this model, these practices are the independent variables that directly influence the dependent variable, the child's nutritional status (measured by stunting, wasting, and underweight). However, this relationship is moderated by intervening variables, including household socio-economic status (income and education) and demographics. These factors can either facilitate or hinder the translation of knowledge into positive nutritional outcomes.

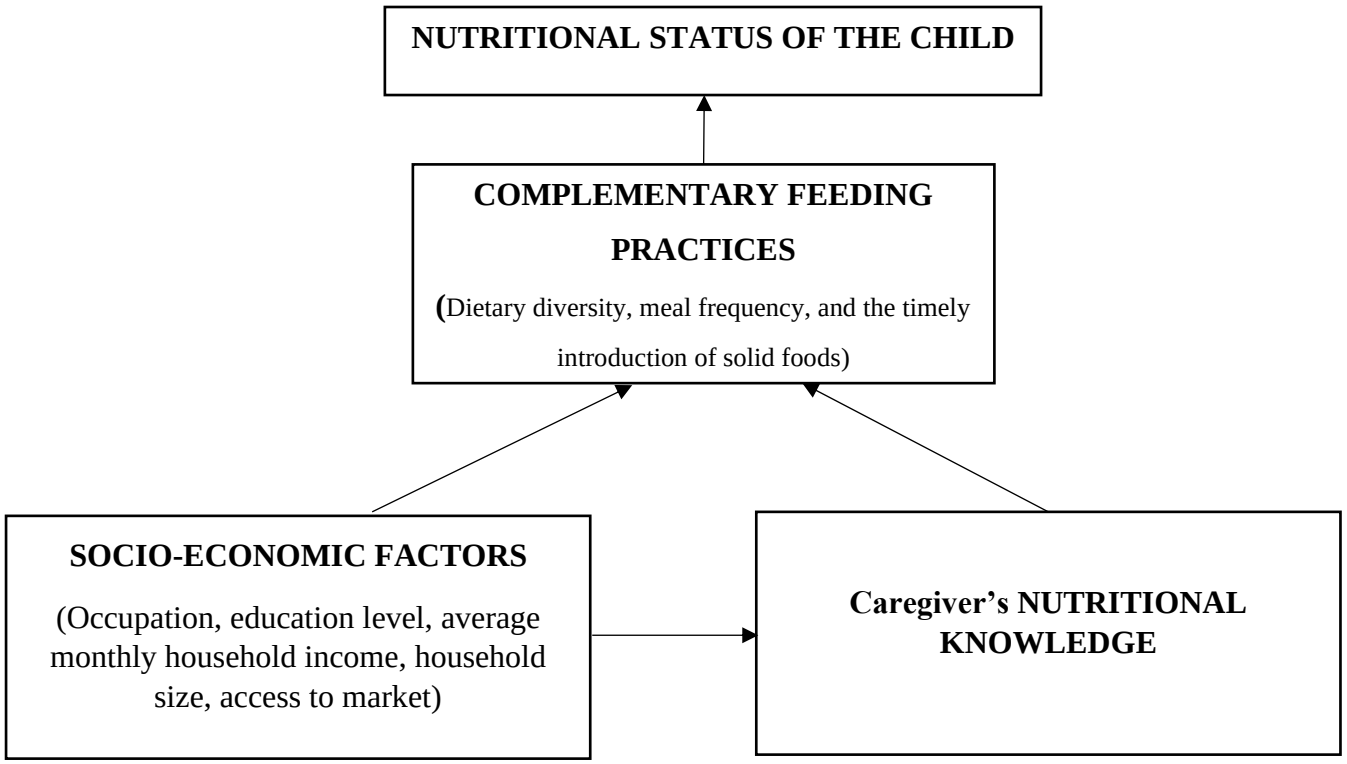


Figure 1: Conceptual Framework for the Study

Source: Author

CHAPTER TWO: LITERATURE REVIEW

2.1 Overview of Complementary Feeding

During infancy and early childhood (birth to 2 years), an adequate amount of appropriate nutrition is paramount for the full development of children's human potential. This period is also regarded as a "critical window" for a child's health, growth, and development (WHO, 2023a). It is a peak period for growth faltering, micronutrient deficiencies, and the emergence of common childhood ailments. Furthermore, reversing stunting developed during this period is very difficult after the child's second birthday (Abeshu et al., 2016).

Conceptually, complementary feeding is defined not merely by the introduction of solids, but by a set of four essential attributes that ensure the transition from breast milk to family foods supports optimal growth:

Timely: This refers to the introduction of complementary foods when an infant's energy and nutrient needs exceed what can be provided by exclusive, frequent breastfeeding. At approximately 6 months of age, an infant's digestive system and kidneys are sufficiently mature to metabolize non-milk foods (Birhanu et al., 2025).

Adequate: To be conceptually sound, the diet must provide sufficient energy, protein, and micronutrients (particularly iron, zinc, and calcium) to meet a growing child's metabolic demands. This is measured through Minimum Meal Frequency (MMF) and Minimum Dietary Diversity (MDD).

Safe: This dimension entails the hygienic preparation and storage of foods to prevent contamination. In urban settings like Kawempe, where water and sanitation challenges exist, safe feeding is a critical barrier against the diarrhea-malnutrition cycle (Murugasen et al., 2024).

Appropriate: This refers to the texture and consistency of the food. Foods must transition from pureed/mashed at 6 months to lumpy/finger foods by 12 months to match the child's neuromuscular development. It also encompasses Responsive Feeding, where caregivers recognize and respond to the child's hunger and satiety cues (WHO, 2023).

Poor nutrition during these formative years has immediate consequences of increased morbidity and mortality, and delayed development of the brain and other nervous systems (Abeshu et al., 2016). The latent impacts include impaired cognitive performance, reduced work capacity, and the persistence of the intergenerational cycle of malnutrition. When a malnourished girl child grows up, she faces greater odds of having a low-birth-weight infant, thereby continuing the cycle of poverty and poor health in low-resource settings (Murugasen et al., 2024).

2.2 Age of Introduction of Complementary Foods

The timely introduction of complementary foods is a non-negotiable pillar of infant health. There is a clear global consensus, led by the WHO and UNICEF, that 6 months (180 days) is the optimal age to begin complementary feeding. This is not an arbitrary figure but is based on the intersection of nutritional necessity and physiological maturation. By 6 months, breast milk alone can no longer meet the surging requirements for energy and micronutrients, particularly iron and zinc, as the infant's birth stores begin to deplete (Dewey & Adu-Afarwuah, 2018).

The infant's kidneys reach a state where they can successfully eliminate the higher renal solute load from proteins found in meat and legumes. Concurrently, the intestinal tract develops a robust defense system and the necessary enzymes to utilize complex fats and carbohydrates (Feyisa et al., 2020). Lastly, at this stage, the extrusion reflex (pushing food out with the tongue) diminishes, and the infant develops the ability to masticate, swallow, and appreciate varying food textures and tastes (Dhami et al., 2019).

The science-based guidelines emphasize that deviation from this window, either early or late, carries significant health risks. Early introduction (before 6 months) increases the risk of choking, food allergies, and the displacement of breast milk, which can lead to early weaning and increased exposure to food-borne pathogens (Fanta & Cherie, 2020). Conversely, delayed introduction misses the window of developmental readiness and leads to rapid growth faltering and stunting (Monte & Giugliani, 2024).

Despite the clarity of these guidelines, studies in similar East African peri-urban and rural settings show high rates of non-compliance. In Ethiopia, a study by Gebretsadik et al. (2023) found that nearly 30% of mothers introduced complementary foods either too early due to the

perception that milk was not enough or too late due to a lack of knowledge on developmental cues.

In Uganda, local evidence from the UDHS (2022) indicates that while the majority of children eventually start complementary foods, the quality of what is introduced at 6 months is often poor, consisting largely of monotonous carbohydrate-based porridges that lack the nutrient density required for this transition. Studies in informal settlements similar to Kawempe highlight that maternal return to work and the influence of aggressive marketing of breast milk substitutes often lead to the premature introduction of transitional foods as early as 3 or 4 months (Dhami et al., 2019).

2.3 Assessment of Complementary Feeding Practices

Assessment of complementary feeding (6–23 months) evaluates the timing, quality, and safety of solid foods introduced alongside breast milk to meet increasing nutritional needs. Key indicators include timely initiation (6 months), minimum dietary diversity, and minimum meal frequency. Poor practices (for example, early initiation, low diversity) are linked to malnutrition, especially in low-income settings (Miniello et al., 2021).

Minimum Dietary Diversity is a proxy for micronutrient adequacy. The WHO now recognizes eight food groups: (1) breastmilk, (2) grains, roots, and tubers, (3) pulses, nuts, and seeds, (4) dairy products, (5) flesh foods, (6) eggs, (7) vitamin A-rich fruits and vegetables, and (8) other fruits and vegetables. Minimum Meal Frequency ensures that the child's energy requirements are met, as their stomach capacity is small. Minimum Acceptable Diet is a composite indicator of both MDD and MMF (WHO, 2023).

2.4 The Complementary Feeding Practices (CP) among Caregivers of children aged 6-23 months

2.4.1 Global and regional trends of Complementary Feeding practices among Caregivers of children aged 6-23 months

Globally, approximately 1 in 2 children are not fed the Minimum Meal Frequency (MMF), and nearly 2 in 3 do not receive the Minimum Dietary Diversity (MDD) required for healthy brain and body development (Barros et al., 2021).

In addition, only about 24% to 30% of children in this age group meet the criteria for a Minimum Acceptable Diet (MAD). As of 2024, 23.2% of children under 5 are stunted, with

more than half of these cases developing during the complementary feeding transition (6–23 months) (WHO,2023).

In West Africa, Issaka et al. (2015) reported MMF rates as low as 42% in Sierra Leone. In Ethiopia, Gebretsadik et al. (2023) and Shukure (2017) identified that while initiation is often timely, the quality of food is frequently compromised by limited variety, often only three food types. Similarly, studies in Tamil Nadu by Kavitha and Nadhiya (2024) showed a preference for commercial weaning foods (64%) due to convenience, despite these being potentially less nutritious than diversified home-cooked meals.

According to (Issaka et al., 2015), many children in developing countries still do not meet the standards for optimal complementary feeding practices. For example, the percentage of children who received the minimum meal frequency was 51% for Ghana and Liberia, 55% for Nigeria, and 42% for Sierra Leone. These low proportions indicate that there might be other factors that pose risks for children to meet the minimum standards f or complementary feeding.

Recent multilevel analyses (2024) indicate that the prevalence of appropriate complementary feeding in SSA is only about 13% (UNICEF, 2021). While East African countries have shown slight improvements in meal frequency, West Africa typically shows higher odds of meeting dietary diversity compared to Central or Southern Africa.

In the study conducted (Shukure, 2017), in Fiche town (Ethiopia), 84% of women initiated complementary feeding of which 61.5% of mothers initiated complementary feeding timely. 23.3% of mothers initiated complementary feeding before the time and the study further revealed the late initiation of complementary feeding was 15.2% and 69.8% of the women feed their child four times and above per day. Among them, 42% of mothers changed breastfeeding frequency after initiation of complementary feeding. 61% of mothers prepared complementary feeding with three types of food.

A similar study further found out that, 40.5% of mothers usually use milk and milk products to prepare complementary food. 38.1% used cup and spoon and 9% used both bottle feeding and cup and spoon. 63.1% of mothers feed their children three times per day, 27.1% feed two times per day. However, 9% feed their children one time per day, while 12.4% of mothers feed their children with a caregiver. Two-thirds of husbands were supported by the economy but 11% had no support.

A rising trend in 2025–2026 is that the Triple Burden is prevalent. The coexistence of stunting, micronutrient deficiencies, and overweight within the same communities. This is often driven by the increased availability of commercially produced complementary foods (CPCF) that are high in sugar and sodium but low in essential nutrients.

2.4.2 Trends of Complementary Feeding practices among Caregivers of children aged 6-23 months in Uganda

As of 2024, approximately 32% of children aged 6–23 months meet the minimum dietary diversity (consuming foods from 5 of the 8 food groups). While this is an improvement from previous years, nearly 68% of Ugandan children are still fed a diet lacking in essential nutrients. Roughly 40% to 45% of children are fed the recommended number of times per day. Frequency is generally higher in urban areas compared to rural regions, and only about 13% to 15% of children receive a Minimum Acceptable Diet.

The UDHS (2022) highlights a stark disparity: while breastfeeding rates are high, the quality and frequency of the solid part of the diet fail to meet the biological needs of a rapidly growing brain and body.

In a study done by Mokori et al. (2017) in Northern Uganda, the proportion of Northern Ugandan children consuming an adequate complementary diet increased by 3.1% between 2006 and 2011, according to their findings. But during this period, the length of breastfeeding increased from 11 to 12 months, and the proportion of mothers who continued to breastfeed their children at two years old dropped from 55.2% to 46.5.

According to a study done by Aber et al. (2018) in Lamwo district, rural Uganda, they found that adherence to complementary feeding guidelines was only 40.1%. The study had several complementary meals ranging from one to four meals per day with a mean of three meals. In line with the above, they found that less than half (42.4%) of the children were reported to have received at least four of the foods from the seven recommended food groups. However, nearly all (97.7%) of the children had ever breastfed and 91.1% received a syrup or capsule for vitamin A supplementation.

A study by Atim et al. (2024) in urban Uganda found that while many children consume grains (staples like matooke or maize), fewer than 20% consume animal-source proteins or vitamin A-rich vegetables daily. This monotonous diet is a primary driver of stunting.

According to a study done by (Asiimwe et al., 2021) in Eastern and Central Uganda which showed that in the 24 hours leading up to data collection, 78% of the children consumed foods from four or more groups, and 83% of children in the 12- to 23-month age group consumed more diverse meals than 70% of children in the 6- to 11-month age group. Additionally, the children aged 6 to 23 months had their intake of foods from each of the different food groups measured. Starchy staples were consumed by 94% of the children surveyed. Nevertheless, the consumption of heme-rich (meat, poultry, organ meats, fish, and shellfish) and vitamin A-rich plant foods, eggs, and milk was utilized to measure the micronutrient intake of the children in this study. According to the results, 68% of children in both regions were fed foods high in vitamin A, 29% were fed milk and milk products and only 11% were fed eggs. Meat and fish consumption were consumed by only 44% of the children.

A study by Nabuuma et al. (2021) conducted in peri-urban Kampala (including areas surrounding Komamboga) found that although dietary diversity (MDD) is often higher in the city due to market access (reaching approximately 48%), the Minimum Meal Frequency (MMF) is frequently lower than in rural areas. This is attributed to maternal employment in the informal sector. Many mothers in Kawempe are market vendors or casual laborers who leave their children with substitute caregivers. This often results in prolonged fasting for the infant or reliance on a single, large morning meal of maize porridge, which lacks nutrient density.

In neighboring Kasubi and Kalerwe, Sserwanja et al. (2022) noted a rising trend in the use of unbranded or locally milled composite flours (often mixtures of soya, maize, and silverfish). While these are intended to be nutritious, the study found that poor hygiene during street-vending and improper storage at the household level led to high rates of diarrheal diseases, which directly undermines the child's nutritional status regardless of the food's initial nutrient content.

Lastly, research specifically focusing on Kawempe’s informal settlements indicates that nearly 90% of caregivers use porridge as the primary complementary food. However, as noted by Wamani et al. (2024) in longitudinal observations of Kampala’s peri-urban centers, this

porridge is often energy-diluted, meaning it is made too thin to pass through a bottle or cup easily, failing to provide the caloric density required for children aged 6–23 months.

2.4 Nutritional status of children 6-23 months

According to the Uganda Demographic and Health Survey (UDHS) 2022, the national average prevalence of stunting is 24.4%, wasting stands at 3.2% and, approximately 9.7% of children are underweight. While national stunting rates have plateaued at 26%, the nutritional status of children in peri-urban areas like Kawempe Division (where Komamboga is located) presents a unique public health paradox (Kakooza et al., 2025). Unlike rural areas, where malnutrition is often linked to seasonal food scarcity, malnutrition in urban Kampala is driven by Hidden Hunger and poor dietary quality.

Studies in informal settlements surrounding Komamboga show that stunting remains high (22-25%), but there is a rising trend of acute malnutrition (wasting) due to frequent diarrheal episodes and poor sanitation (Hamid Namaganda et al., 2023). A critical issue in this region is the lack of micronutrients. Even children who are not underweight by traditional scales often suffer from Anemia (affecting over 50% of children in Kampala) because their diets are dominated by starches like matooke and refined maize flour without animal-source proteins (Atim et al., 2024).

Currently, while we know the general statistics for Kampala, there is a lack of site-specific data for Komamboga Health Centre III regarding how the specific employment patterns of urban mothers (market vending, casual labor) directly translate into the Z-scores of their children. This study seeks to fill that gap by linking actual feeding frequencies to the current nutritional status of children visiting this specific facility.

2.5 Nutritional Knowledge of Caregivers regarding Complementary Feeding

According to a study done by El et al., (2014), in Wakiso district, Uganda, about 5.0% of the mothers planned to introduce their first solid foods to their infants at 0-1months, 22% at 2-3 months, and 5.0% at 4-5 months. Moreover, 30% of all the mothers planned to give mostly beans as the first solid food at 0-1 months, 62% soy porridge at 2-3 months, and 60% beans at 4-5 months. However, the mother's age was not significant in determining the age for the introduction of first solid foods fed to the infants but insufficient breast milk (69%) was the main reason given for the early introduction of solid foods, 10% work, and 4% was due to advice. Similarly, 99% of all mothers agreed that more information regarding the introduction

of infant solid foods should be made available with 39% of those preferring it through clinical nurses and 21% health practitioners.

A study conducted by (Achiro et al., 2023), in Northern Uganda, found that there were varied ages of the introduction of complementary feeding and more than half (54.3%) of the caregivers did not adhere to the WHO and MOH of Uganda guidelines for introducing complementary feeding at 6 months, while less than half (45.7%) adhered to the guideline.

In a study done by (Asiimwe et al., 2021), in Eastern and Central Uganda, it discovered that 77% of caregivers were knowledgeable about a variety of topics, including the appropriate age to begin complementary feeding and ingredients that can be added to improve a baby's food's nutritional content (94%). However, of the topics covered in the survey, a relatively high percentage of caregivers (26%) were not aware of ways to encourage children to eat more food. Some claimed that making a child eat would increase their intake, while others claimed that adding sweetness to food would increase a child's appetite. These practices are not recommended because they could lead a child to associate food with discomfort and suffering, while the latter is linked to health.

Another study done in the Konaseema region by (Sethi et al., 2017), discovered that 42.0% of mothers knew that complementary feeding should begin by six to seven months, 26.8% by eight to nine months, 20.0% by ten to twelve months, 1.2% by more than twelve months, and 4.2% had no idea what complementary feeding was. Additionally, there was very little information available about the consistency, quantity or amount, and kind of complementary foods that should be started.

According to a study done in Fiche town in Ethiopia (Shukure, 2017), 64% of mothers had access to information about complementary feeding. Out of the mothers who had access to information about complementary feeding, 62.5% of women obtained the information from the health institution and 52.5% got the information from health extension workers. From this 18.75% of mothers got the information weekly, others 18.75%, and 62.5% got the information monthly and when they went to health institutions. Meanwhile, 61.2% answered the proper time for starting complementary feeding at six months, others, immediately after birth,1-3 months, 4-6 months, -12 months, and above 12 months are 7.2%, 5.2%, and 10.8%. 8.4% and 7.2%respectively.

In a study done in Kosova by (Berisha et al., 2017), 96% of mothers' knowledge shows significant distinction according to higher level of education regarding perceptions that complementary feeding is the act of giving foods and liquids other than breast milk to the infants, and 61.9% knew that complementary food should be given to the infants after 6 months both high school and university.

2.6 Research gaps

While there is substantial literature on Infant and Young Child Feeding (IYCF) in Uganda, several critical research voids persist. These gaps are not merely limited to a lack of caregiver awareness but represent systemic gaps in the existing body of scientific literature that this study intends to fill.

2.6.1 Limited Research on Peri-Urban Informal Health Systems

Most nutritional research in Uganda is either national in scope, such as the UDHS (2022), or focused on high-burden rural regions like Karamoja. There is a dearth of primary research focused on the unique dynamics of HC III-level facilities within densely populated, "slum-characterized" peri-urban environments. While the findings are centered on Komamboga HC III, the study addresses a broader gap: how primary healthcare systems in informal urban settlements manage the transition from exclusive breastfeeding to complementary feeding.

2.6.2 The Urban Care Paradox and Maternal Employment

While it is well documented that urban areas have better access to food markets, recent evidence suggests an urban care paradox where maternal employment in the informal sector leads to the delegation of feeding to siblings or untrained neighbors (Mekonen et al., 2024). There is limited evidence on how these specific care arrangements influence the Minimum Dietary Diversity (MDD), Minimum Meal Frequency (MMF), and Minimum Acceptable Diet (MAD). Most studies assume the mother is the primary feeder, failing to account for the proxy-caregiver effect common in Kampala's peri-urban slums.

2.6.3 Socio-Behavioural and Economic Determinants

Current literature often treats poverty as the sole driver of malnutrition. However, some studies have shown that even in households with adequate resources, children still suffer from stunting due to cultural myths such as the belief that eggs cause speech delays or that fish lead to worms (MOH, 2021). There is a gap in understanding the interaction between economic constraints and these deep-seated cultural behavioral drivers within the diverse, multi-ethnic population of Kawempe.

2.6.4 Emerging Hidden Hunger in Urban Settings

While stunting and wasting are frequently measured, the specific hidden hunger or micronutrient gaps (such as Vitamin A and Iron) among children 6-23 months in informal settlements in Kampala are infrequently tracked alongside anthropometry (Gain et al., 2020). This study bridges the gap by linking specific dietary diversity indicators directly to the physical nutritional status of children attending the health centre.

CHAPTER THREE: METHODOLOGY

3.1 Study Design

The study employed a facility-based, cross-sectional study design and a mixed-methods approach. This design was appropriate as it allows for the collection of data on caregiver practices and nutritional status at a single point in time to determine the prevalence of malnutrition and its associated feeding factors (Creswell & Creswell, 2023).

3.2 Study Area

The study was conducted at Komamboga Health Centre III, situated in the Kawempe Division of Kampala District, approximately 8 kilometers north of the city center. It is strategically located along the Kampala-Gayaza Road, making it a primary access point for the rapidly growing peri-urban populations of Komamboga, Kyanja, and Mpererwe. The facility sits within a high-density area characterized by informal settlements and low-income households, which are the primary focus of the nutritional assessment.

Komamboga HC III operates under the Kampala Capital City Authority (KCCA) health department. Within the regional referral network, it serves as a critical buffer for Kawempe National Referral Hospital and Mulago National Referral Hospital. It typically receives patients from lower-level private clinics and community health posts while referring complicated pediatric and obstetric cases to the national referral units.

The facility is organized into several specialized units, most notably the Young Child Clinic (YCC) and the Antenatal Clinic (ANC). The YCC is the primary site for the study, as it conducts routine Growth Monitoring and Promotion (GMP), nutritional surveillance and immunization services

In addition to basic care, Komamboga HC III hosts the Integrated Management of Childhood Illness (IMCI) program and provides specialized outpatient therapeutic care for children identified with moderate acute malnutrition. Its high patient volume and integration of nutritional education within the pediatric department make it an ideal environment for observing the practical application of caregiver knowledge.

3.3 Target Population

The target population consisted of all caregivers (mothers, fathers, or guardians) with children aged 6-23 months who attend the Komamboga Health Centre III for healthcare services.

3.4 Inclusion and Exclusion Criteria

3.4.1 Inclusion Criteria

i. Caregivers with children aged 6–23 months attending Komamboga HC III during the study period.
ii. Caregivers who have lived in Kawempe Division for at least six months to ensure localized urban context.
iii. Caregivers who provide informed consent to participate.

3.4.2 Exclusion Criteria

i. Children with known congenital malformations or chronic illnesses (e.g., HIV/AIDS, TB) that independently affect nutritional status.
ii. Caregivers whose children need emergency medical stabilization at the time of the visit

3.5 Sample size determinations

The Cochran equation was used to calculate the sample size.

$N = Z^2 pq/d^2$

Where;

N = sample size

Z is a constant =1.96 at the 95% Confidence Interval.

P = the estimated prevalence (Estimated prevalence of stunting in Kampala, 16.5%, based on UDHS 2022, UBOS, 2023).

q = 1-p

d = 5% confidence, which is 0.05

$N = (1.96)^2 * 0.165 * (1-0.165) / (0.05)^2$

N = 212

5% non respondence rate was considered

Therefore: 5% of 212 = 10

N = 212+10 = 221

221 respondents

However, due to limited resources (financial and human resources), a sample size of 170 participants was considered.

3.6 Sampling Techniques

The study employed a multistage sampling technique to ensure the sample was representative of the diverse socio-economic backgrounds within the Kawempe Division.

3.6.1 Purposive sampling

Komamboga Health Centre III was purposively selected as the primary study site. As a key public health facility in Kawempe, it served as a central point for maternal and child health (MCH) services for a vast catchment area.

Five Key Informants (1 nutritionist, 2 nurse, 1 clinical officer, and 1 VHT) were selected using purposive sampling to provide in-depth insights into specific aspects of complementary feeding practices.

3.6.2 Systematic Random Sampling

On each scheduled clinic day, the following steps were taken to select respondents. A list of all eligible caregivers was generated from the morning registration desk or the Young Child Clinic (YCC) register as they arrive. The sampling interval (k) was determined by dividing the expected daily attendance by the required daily quota (k = 300/150). According to the data clerk at the facility, the monthly ANC attendance by pregnant mothers is 300. A random starting point between 1 and k was chosen using a random number table or a draw. Every 2nd caregiver thereafter was approached for recruitment until the daily quota for that specific stratum was achieved.

3.7 Data collection methods

3.7.1 Face-to-face interview

Data was collected through individual, face-to-face interviews conducted by trained research assistants at Komamboga Health Centre III. Once a caregiver was identified through the systematic random sampling process, he/she was approached.

The interview was conducted in the language the caregiver is most comfortable with, primarily Luganda or English, to ensure there were no communication barriers. The interview was conducted in a quiet, semi-private area within the health Centre to ensure the caregiver feels comfortable sharing household information.

3.7.2 Anthropometric Measurements

All measurements were conducted in accordance with the WHO Multicentre Growth Reference Study protocols (WHO, 2006) to ensure international comparability and technical accuracy. To take the weight measurement, a calibrated digital SECA scale (to the nearest 0.1 kg) was used. The scale was placed on a flat, hard surface and tared (zeroed) before each use. Children were weighed with minimal clothing. For children who can stand steadily, they were placed in the center of the scale. For younger or distressed infants, the tared weighing (Mother-and-Child) method were used, the caregiver was weighed first, the scale was tared, and then the caregiver was handed the child to obtain the child's weight alone.

To take the height measurement, a standard wooden or plastic infantometer (length board) with a fixed headboard and a sliding footpiece (to the nearest 0.1 cm). Since all subjects were under 24 months, recumbent length was measured while the child was lying down. Two trained researchers were required. One to hold the child's head against the fixed headboard and another to straighten the child's legs and press the sliding footpiece firmly against the heels.

Lastly, for MUAC measurement, a standard, non-stretch, color-coded UNICEF MUAC tape. The measurement was taken on the left arm (less active arm). We identified the midpoint between the tip of the shoulder (acromion) and the tip of the elbow (olecranon). The tape was wrapped around this midpoint, neither too tight nor too loose. Results were recorded in centimeters and categorized by color: green (Well-nourished), yellow (At risk/Moderate Acute Malnutrition), and red (Severe Acute Malnutrition).

The presence of nutritional edema was also assessed. We applied firm thumb pressure for three seconds on the tops of both feet. If a pit (indentation) remains after the thumbs are removed, bilateral pitting oedema was recorded, indicating a severe form of malnutrition.

3.7.3 Key Informant Interviews

Key Informant Interviews were conducted to obtain in-depth qualitative insights into the systemic and community-level factors influencing child-feeding practices in Kawempe Division. A total of 5 purposively selected informants were interviewed. These included 2 nutritionists, a nurse, a clinical officer, and a VHT, to gather information on the context of child-feeding practices and the factors that influence them. The KIIs was conducted using a semi-structured interview guide, allowing for flexible yet focused discussions. This qualitative data was used to triangulate the quantitative findings from the caregiver questionnaires.

3.8 Data Collection Tools

3.8.1 Structured Questionnaire

The primary tool for the face-to-face interview was a structured, researcher-administered questionnaire. The questionnaire (See Annex 2) was organized into the following standardized modules as socio-demographic characteristics, 24-Hour Dietary Recall, IYCF Practices Scorecard, and caregiver knowledge assessment.

3.8.2 Anthropometric Tools

An electronic SECCA weighing scale was used to obtain the child's weight. This specific model features a 2-in-1 or tared function, which allows infants to be weighed accurately while held in the caregiver's arms by automatically subtracting the caregiver's weight.

A portable wooden infantometer was used to measure the recumbent length of children under the age of two. The board included a fixed headpiece and a sliding footpiece to ensure accuracy to the nearest 0.1 cm.

Standardized, color-coded UNICEF tapes were used to measure the Mid-Upper Arm Circumference (MUAC). These tapes provided a rapid screening method for acute malnutrition by categorizing the child's nutritional status based on established mid-arm muscle circumference thresholds.

3.8.3 Key Informant Interview Guides

A semi-structured guide with open-ended questions targeting the nutritionists, nurses, clinical officers, and VHT was used (See Annex 3). It focused on barriers to implementing appropriate complementary feeding practices at Komamboga Health Centre III.

3.9 Data Quality Assurance

3.9.1 Reliability

Completed questionnaires were checked daily for accuracy and completeness in the recording of responses. Editing and coding were done before data entry. The Cronbach's alpha coefficient was used to assess internal consistency (reliability), and a coefficient value >0.82 was obtained which supported the internal consistency.

3.9.2 Validity

Content validity was established through the review of the research tools by my supervisor at the University. His expertise helped to determine if the instrument adequately covers all relevant aspects of the study's objectives.

3.9.3 Pretesting

Before the main data collection begas, a pilot study was conducted with 10% (15 participants) of the sample size. The pre-test aimed to identify ambiguous questions and estimate the actual time taken per interview. Feedback from the pilot study were used to refine the wording of the questions and adjust the visual aids before the final rollout at Komamboga Health Centre III. Data from the pilot study were not included in the final analysis.

3.10 Data Analysis

3.10.1 Analysis of Nutritional Knowledge of the Caregivers

To evaluate the nutritional knowledge of caregivers, a structured scoring system was employed to transform qualitative responses into a quantitative index.

Knowledge was assessed using 10 closed-ended and multiple-choice questions adapted from the FAO/WHO Guidelines for Assessing Nutrition-Related Knowledge. Each response matching the standard guideline was coded as 1 (Correct), while incorrect answers, misconceptions, or don't know responses were coded as 0 (Incorrect).

The aggregate scores were categorized into three levels of proficiency based on a modified Bloom's cut-off point. High Knowledge was defined by scores of 8–10, Moderate Knowledge was defined by scores of 6–7, Low Knowledge was defined by scores <6.

3.10.2 Analysis of Complementary Feeding Practices

Complementary feeding practices were analyzed using three core WHO/UNICEF indicators: Minimum Dietary Diversity (MDD), Minimum Meal Frequency (MMF), and Minimum Acceptable Diet (MAD).

The analysis of MDD served as a proxy for the micronutrient adequacy of the child's diet. Data from the 24-hour dietary recall were coded into a binary format (1 = Consumed, 0 = Not Consumed) for each of the following eight groups: breastmilk, grains, roots and tubers, pulses, nuts and seeds, dairy products, flesh foods, eggs, Vitamin A-rich fruits and vegetables, and other fruits and vegetables. Dietary Diversity Score was created by summing the 'Yes' responses across these eight food groups.

A Dietary Diversity Score was calculated for each child by summing the affirmative responses across these eight groups, resulting in a score ranging from 0 to 8. Following WHO guidelines, the nutritional adequacy of the diet were interpreted categorically: children who consume five or more of these food groups were classified as having achieved Adequate Dietary Diversity, while those consuming fewer than five groups were classified as having Inadequate Dietary Diversity.

Minimum Meal Frequency was analyzed based on the child's age and breastfeeding status. For Breastfed Infants (6-8 months): The child must have consumed solid, semi-solid, or soft foods at least two times in the previous 24 hours. For Breastfed Children (9-23 months): The child must have consumed solid, semi-solid, or soft foods at least three times in the previous 24 hours. For Non-Breastfed Children (6-23 months): The child must have consumed solid, semi-solid, or soft foods or milk feeds (including formula and animal milk) at least four times in the previous 24 hours.

Minimum Acceptable Diet (MAD): This composite indicator was calculated as the proportion of children who met both the Minimum Dietary Diversity and the Minimum Meal Frequency requirements for their age group.

3.10.3 Analysis of Nutritional Status of Children 6-23 months

To determine the child's nutritional status, the raw data (age, sex, weight, and height/length) were processed using the WHO Anthro software to generate Z-scores based on the 2006 WHO Child Growth Standards.

Weight-for-Height Z-score (WHZ) were generated to determine the prevalence of wasting (acute malnutrition), while Height-for-Age Z-score (HAZ) was generated to determine the prevalence of stunting (chronic malnutrition). Finally, the Weight-for-Age Z-score (WAZ) was generated to determine the prevalence of underweight. Children with a Z-score below -2 were classified as malnourished, and those below -3 SD as severely malnourished.

3.10.4 Statistical Analysis

Descriptive statistics, such as frequencies and percentages, were used to describe socio-demographics. Means and standard deviations (SD) were calculated for continuous variables. Chi-square Test was used to test the association between complementary feeding practices, nutritional knowledge, and nutritional status.

3.10.5 Qualitative Data Analysis

Qualitative data from Key Informant Interviews (KIIs) wasanalyzed using Thematic Content Analysis. Audio recordings were transcribed verbatim and translated into English. Data was organized into codes representing recurring ideas. Related codes were grouped into broader themes to explain the quantitative results.

3.10.6 Data Triangulation

The findings from the quantitative study and anthropometric measurements were compared and contrasted with the qualitative themes. This mixed-methods triangulation provided a comprehensive conclusion on how complementary feeding practices directly impact the nutritional survival of children aged 6-23 months.

3.11 Ethical Considerations

An introductory letter was obtained from the Department of Human Nutrition and Dietetics at Kyambogo University, signed by the research supervisor. This letter served as formal evidence that the research proposal has been reviewed and approved for academic purposes (See Annex 4). It was presented to the In-charge of Komamboga Health Centre III and the Kawempe

Division Medical Officer to seek administrative clearance and introduction to the relevant clinic departments. Lastly, at Komamboga Health Centre III, authorization was sought from the health centre management.

CHAPTER FOUR: RESULTS

4.1 Socio-demographic Characteristics of the Participants

As detailed in Table 1, Primary caregiving in the study population was majorly provided by biological mothers, with grandmothers and other extended family guardians accounting for minor proportions of the primary caregiving responsibilities. Regarding child demographics, female children represented nearly two-thirds of the evaluated sample. The overall mean age among caregivers was 26.8 (5.2) years. The majority of caregivers were married or cohabiting. In terms of household composition, more than half of the study households were medium-sized, followed by large families containing seven or more members, and small households comprising three members or fewer.

Educational levels were mainly centered around secondary-level schooling, followed by primary-level education or below, while caregivers with tertiary qualifications formed the smallest educational subgroup. Economic activity was characterized by informal and vulnerable livelihoods; informal casual labor emerged as the primary source of household income, followed by small-scale commercial trade. Formal salaried employment and agricultural farming represented minor segments of the working population.

Table 1: Socio-demographic Characteristics of the Participants (N=170)

Variable Category	Characteristic	Frequency (n)	Percentage (%)
Caregiver Relationship	Biological Mother	151	88.8
	Grandmother	13	7.6
	Other Primary Caregiver	6	3.5
Age Group (Years)	<20	18	10.6
	20−29	74	43.5
	30−39	46	27.1
	40−49	21	12.4
	≥50	11	6.5
Educational Level	Primary Education or Below	64	37.6
	Secondary Education	78	45.9
	Tertiary Education	28	16.5
Primary Occupation	Unemployed / Housewife	31	18.2
	Casual Laborer	63	37.7
	Farming	12	7.1
	Small Retail	40	23.5
	Formal Employment / Other	16	9.4
Main Income Source	Partner Support	75	44.1
	Business / Self-Employment	35	20.6
	Formal Wages	20	11.8
Household Size	Small (1−3 members)	36	21.2
	Medium (4−6 members)	91	53.5
	Large (≥7 members)	43	25.3

4.2 Nutritional Knowledge Regarding Complementary Feeding among Caregivers

4.2.1 Distribution of Caregivers' Nutritional Knowledge Levels

As illustrated in Figure 2, nearly half of the surveyed caregivers demonstrated a moderate level of nutritional knowledge. About one-quarter of the participants exhibited a high level of knowledge, while a quarter fell into the low knowledge category. Overall, the mean knowledge score achieved across all caregivers was 61.4% (14.8).

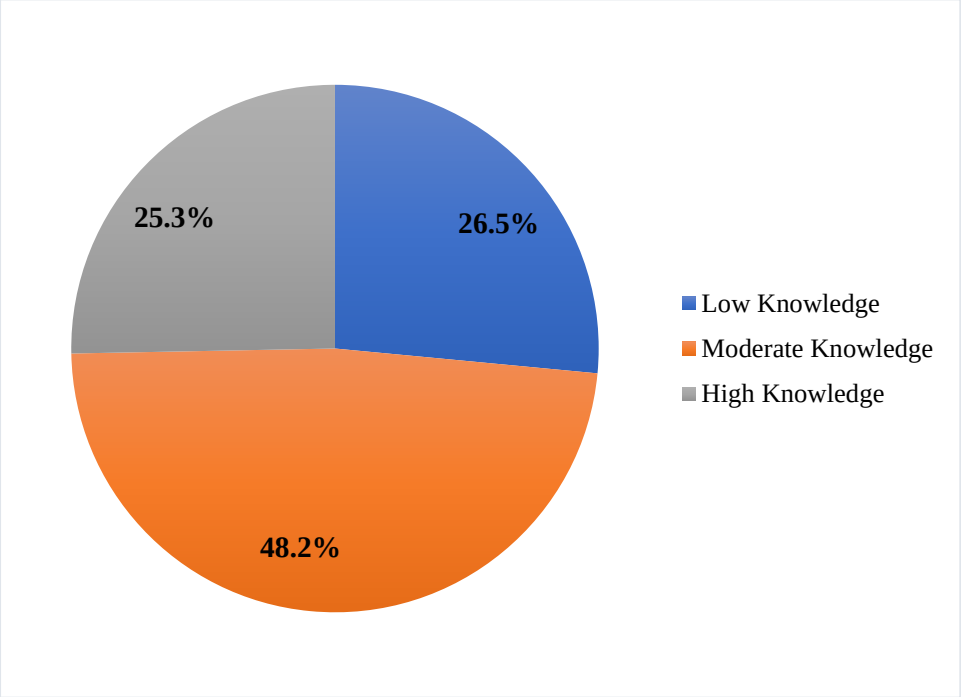


Figure 2: Distribution of Caregivers' Nutritional Knowledge Levels

4.2.2 Knowledge about various practices

A high proportion of caregivers correctly identified 6 months as the recommended age for introducing complementary foods alongside continued breastfeeding (Figure 3). However, knowledge regarding age-appropriate meal frequency was less consistent. More than half correctly noted that young children require meals 3 to 4 times daily, plus 1 - 2 nutritious snacks. Majority recognized iron-rich animal-source foods as essential for child growth and cognitive development and identified orange-fleshed sweet potatoes, pumpkins, and dark green leafy vegetables as crucial for immunity and vision.

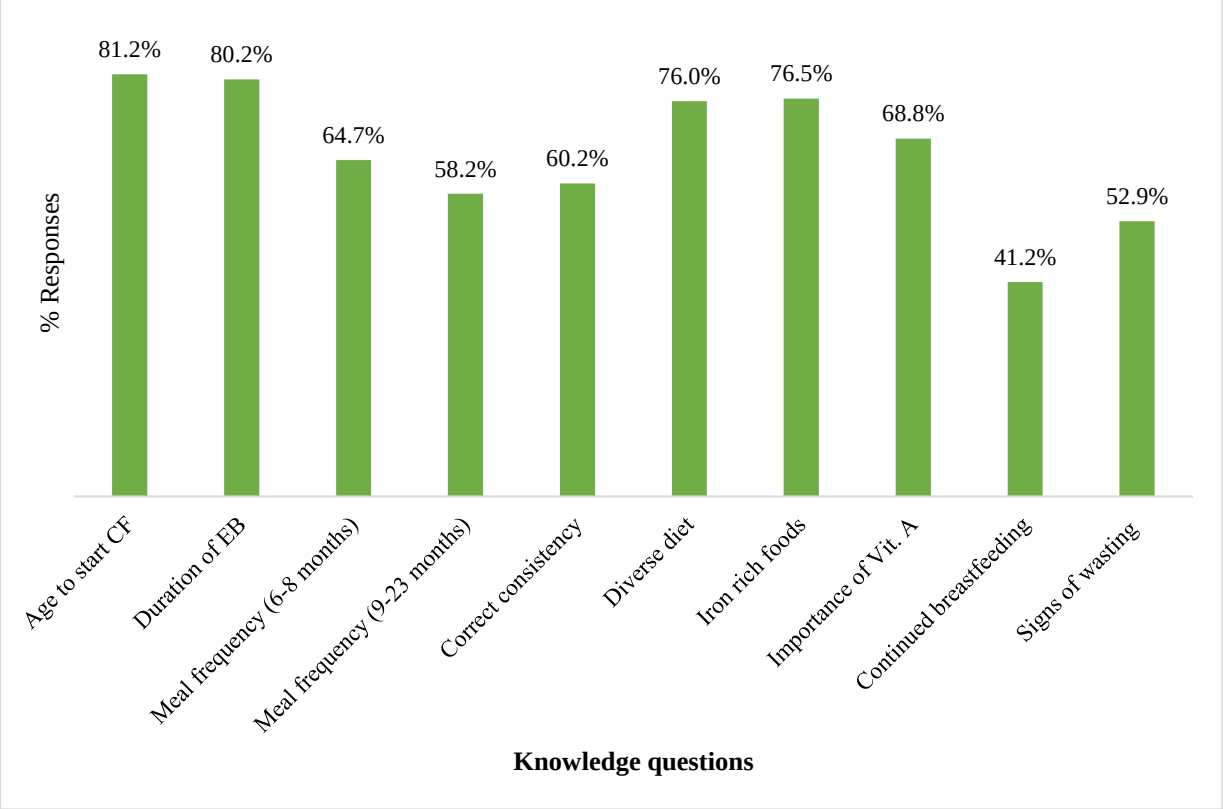


Figure 3: Knowledge about various practices

4.3 Complementary Feeding Practices among Caregivers of Children aged 6-23 months

4.3.1 Minimum Acceptable Diet

As shown in Figure 4, Overall, minimum meal frequency emerged as the most widely achieved practice among the surveyed children, whereas dietary diversity posed a far greater compliance challenge. As a result of this low diversity, under a third of the total child population successfully attained a Minimum Acceptable Diet. Majority of breastfeeding children achieved the age-appropriate meal frequency threshold required for breastfed infants. However, dietary diversity remained low, with under two-fifths receiving foods from the recommended minimum number of food groups. Consequently, less than a third of breastfed infants achieved a Minimum Acceptable Diet.

Among non-breastfeeding children, compliance across all IYCF metrics was visibly lower compared to their breastfed peers. Less than half of the non-breastfed children met the higher meal frequency required for non-breastfed infants, and just over a third attained minimum dietary diversity. When factoring in the additional requirement of at least two dairy/milk feeds alongside diversity and frequency, less than a quarter of non-breastfed children achieved a Minimum Acceptable Diet.

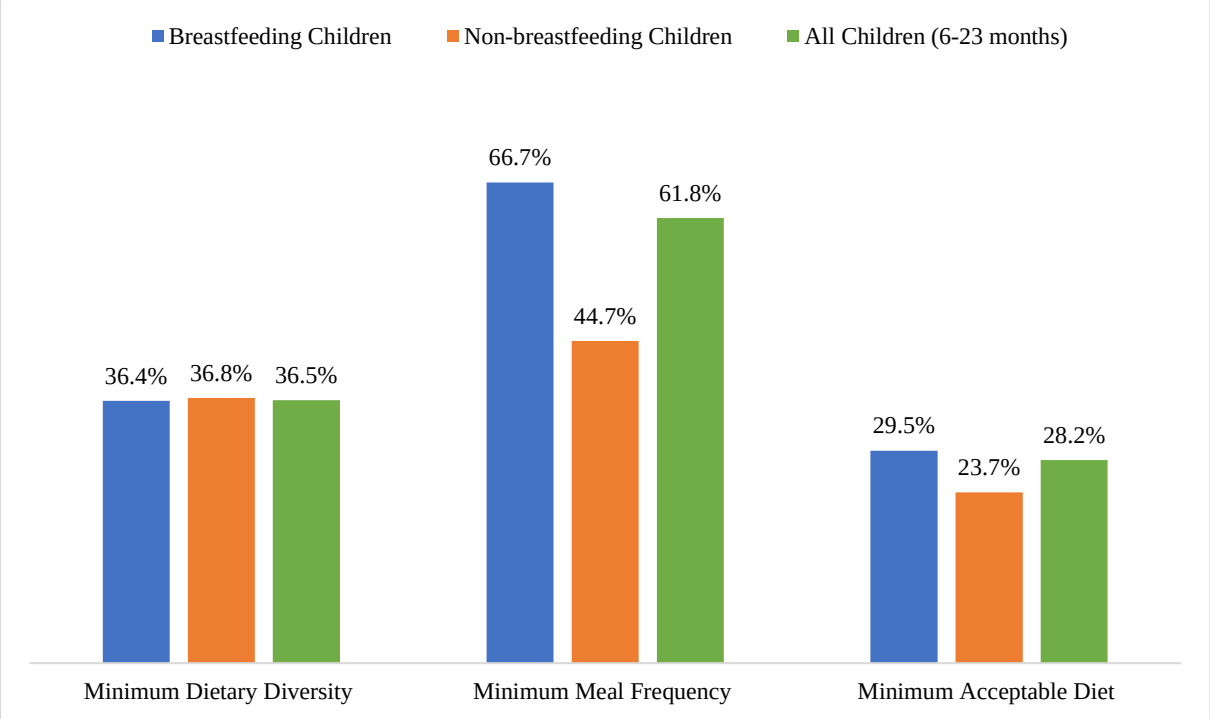


Figure 4: IYCF practices among children 6-23 months

4.3.2 Consumption of various food groups

Starchy foods formed the foundation of complementary diets for nearly all children in this study (Figure 5). Grains, Roots, and Tubers were consumed by most of children, primarily in the form of maize or millet porridges, mashed matooke, sweet potatoes, and cassava. Over three-quarters of the children continued to receive breastmilk alongside complementary foods during the 24-hour recall period. Over half of the children consumed Vitamin A-rich produce, mainly pumpkin, dark green leafy vegetables, and carrots incorporated into family dishes. Pulses, Nuts, and Seeds were consumed by less than half of children, predominantly through groundnut (peanut) sauce or beans mixed into porridge. Other Fruits and Vegetables were consumed by moderate proportion of children, mainly consisting of ripe bananas, pawpaw, and avocados.

Animal-source foods were the least frequently consumed food groups. Dairy Products were consumed by a quarter of children, mainly as small amounts of cow's milk added to tea or porridge. Only few consumed meat, poultry, or fish including powdered silver fish/mukene. Eggs comprised the lowest dietary intake across all groups, consumed by very few children.

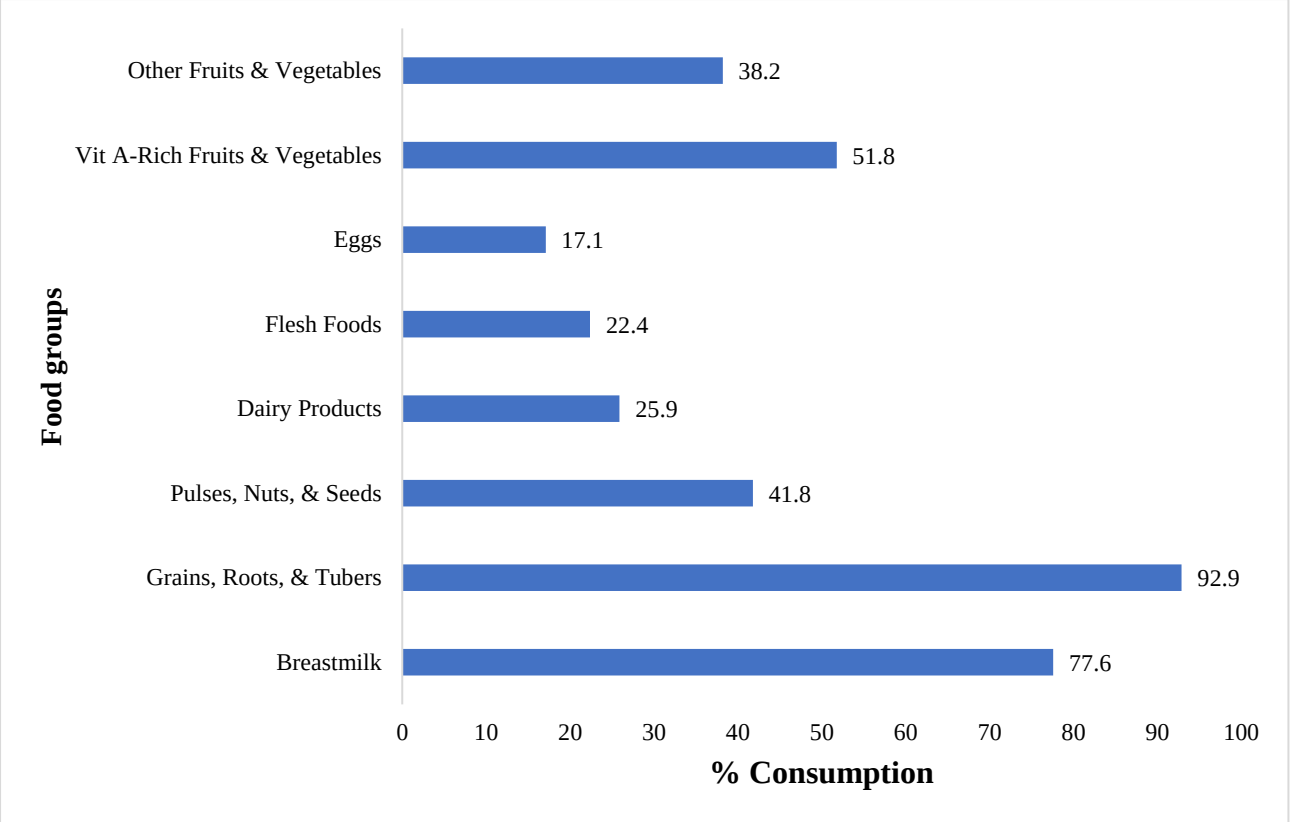


Figure 5: Consumption of various food groups

4.4 The Nutritional Status of Children aged 6-23 months

4.4.1 Prevalence of Wasting

Acute malnutrition (wasting), reflecting short-term weight loss and acute tissue depletion, was observed in a moderate proportion of the children. The majority of affected infants presented with moderate acute wasting, while a smaller fraction exhibited severe acute malnutrition. Male children exhibited a slightly higher rate of acute malnutrition compared to female children.

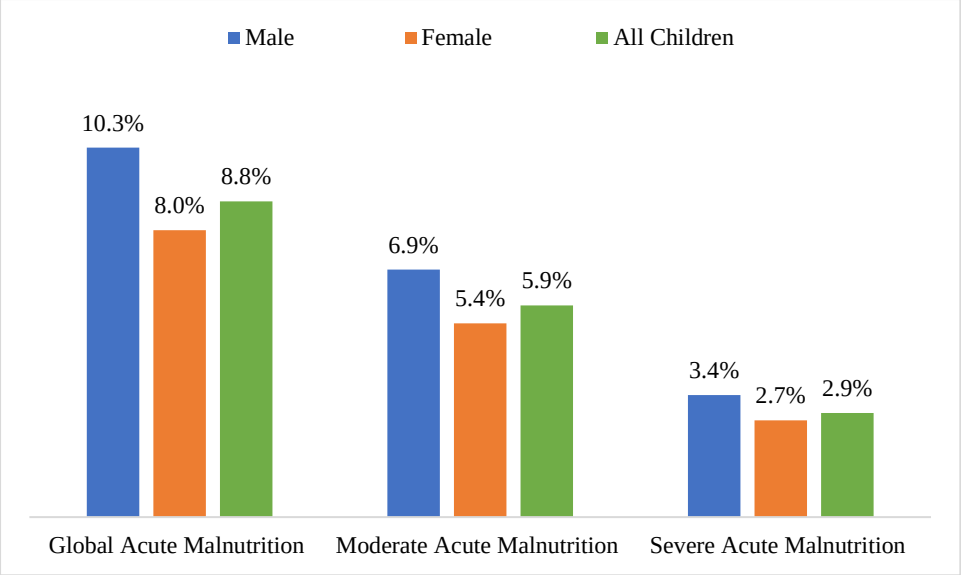


Figure 6: Prevalence of wasting

4.4.2 Prevalence of Stunting

Linear growth faltering (stunting) emerged as the dominant form of undernutrition in the study population. According to WHO public health severity thresholds, the observed prevalence of stunting categorizes the study area as experiencing a high public health concern. Severe chronic growth faltering accounted for a notable sub-segment of these affected infants. Male children exhibited a slightly higher vulnerability to linear growth failure than female children.

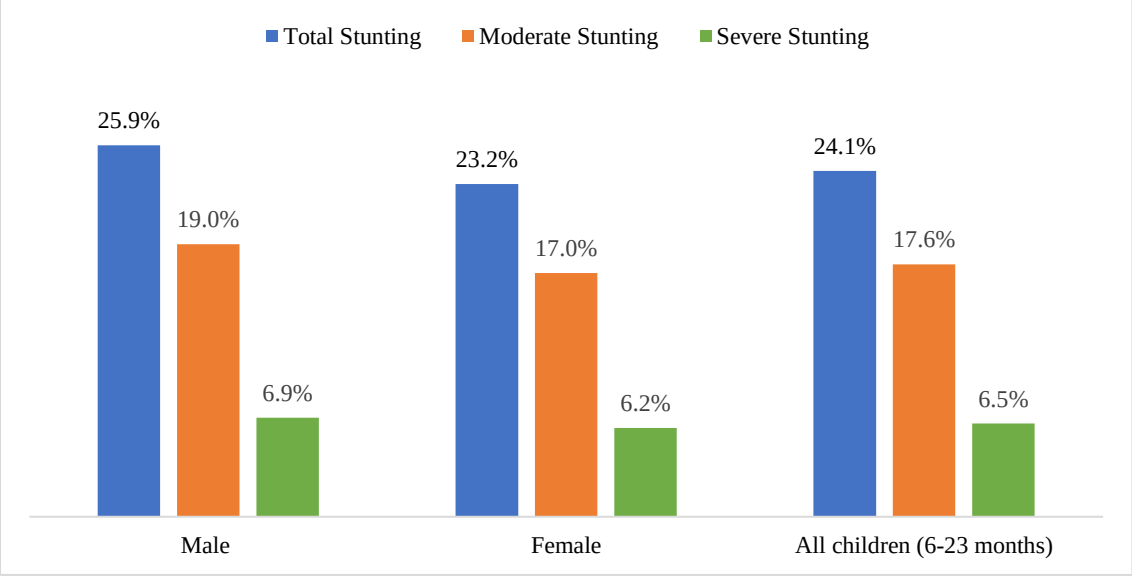


Figure 7: Prevalence of stunting

4.4.3 Prevalence of Underweight

Underweight, which integrates both chronic (stunting) and acute (wasting) nutritional deficits, affected a substantial proportion of children, placing the study setting in the medium public health concern category.

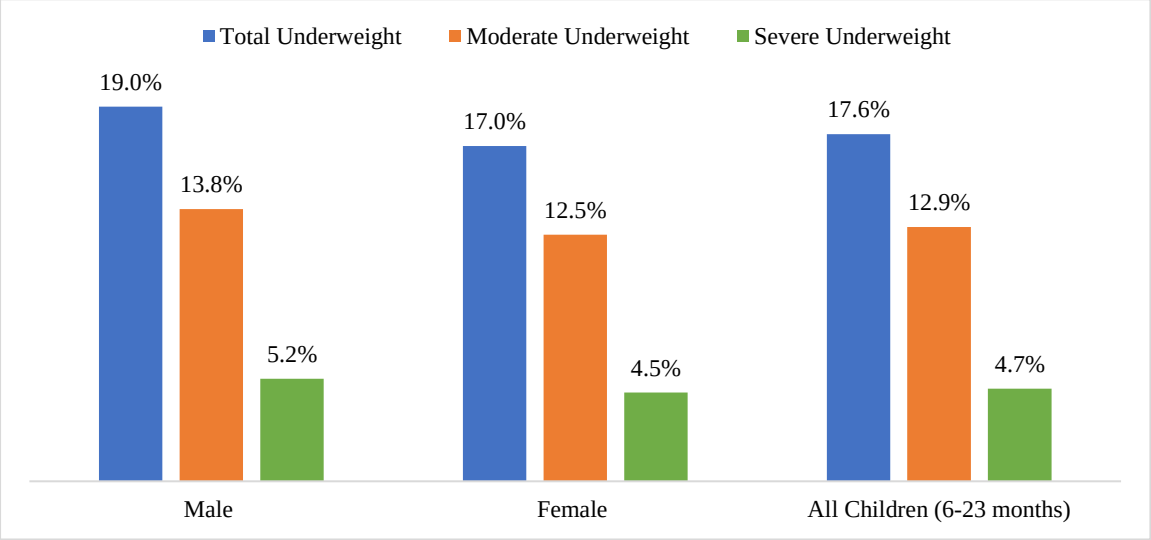


Figure 8: Prevalence of Underweight

4.5 Relationship between caregiver complementary feeding practices and the nutritional status of children aged 6-23 months

A statistically significant association was observed between MAD achievement and stunting ($\chi 2 = 5.857$, $p = 0.015$). Specifically, children who achieved a Minimum Acceptable Diet had a significantly lower rate of stunting compared to those who failed to meet the indicator. Children who failed to achieve a Minimum Acceptable Diet were nearly three times as likely to be stunted compared to those who achieved MAD.

Inadequate dietary diversity was significantly associated with higher rates of stunting ($\chi 2 = 3.892$, $p = 0.049$). Children consuming less than 5 food groups were significantly more likely to be stunted compared to those who consumed diverse diets. No statistically significant associations were found between feeding indicators and acute wasting ($p > 0.05$).

Table 2: Relationship between caregiver complementary feeding practices and the nutritional status (N=170)

Indicator	Category	Stunting (HAZ)		Wasting (WHZ)		Underweight (WAZ)	
		Stunted (<−2 SD)	Normal (≥−2 SD)	Wasted (<−2 SD)	Normal (≥−2 SD)	Underweight (<−2 SD)	Normal (≥−2 SD)
Minimum Dietary Diversity (MDD)	Achieved	7 (14.6%)	41 (85.4%)	3 (6.3%)	45 (93.7%)	5 (10.4%)	43 (89.6%)
	Not Achieved	34 (27.9%)	88 (72.1%)	12 (9.8%)	110 (90.2%)	25 (20.5%)	97 (79.5%)
	Test Statistic ($\chi 2$)	3.892		0.584		2.451	
	p-value	0.049*		0.445		0.117	
Minimum Meal Frequency (MMF)	Achieved	14 (18.4%)	62 (81.6%)	5 (6.6%)	71 (93.4%)	9 (11.8%)	67 (88.2%)
	Not Achieved	27 (28.7%)	67 (71.3%)	10 (10.6%)	84 (89.4%)	21 (22.3%)	73 (77.7%)
	Test Statistic ($\chi 2$)	2.428		0.812		3.21	
	p-value	0.119		0.368		0.073	
Minimum Acceptable Diet (MAD)	Achieved	5 (10.4%)	43 (89.6%)	2 (4.2%)	46 (95.8%)	4 (8.3%)	44 (91.7%)
	Not Achieved	36 (29.5%)	86 (70.5%)	13 (10.7%)	109 (89.3%)	26 (21.3%)	96 (78.7%)
	Test Statistic ($\chi 2$)	5.857		1.087		3.149	
	p-value	0.015*		0.297		0.076	

**Statistically significant at $p < 0.05$.*

4.6 Healthcare Workers' Perspectives on Infant and Young Child Feeding (IYCF) Practices

4.6.1 Health System Capacity and Nutrition Education Delivery

Healthcare workers evaluated the facility's current capacity to deliver nutrition education and reflected on the structural constraints hindering effective complementary feeding counseling during routine child health contacts.

Healthcare workers noted that formal nutrition education is predominantly integrated into routine Antenatal Care (ANC) clinics, Expanded Programme on Immunization (EPI) sessions, and postnatal Growth Monitoring and Promotion (GMP) contacts. Under standard national guidelines for Infant and Young Child Feeding (IYCF), nutrition counseling is intended to be context-specific and tailored, meaning healthcare providers assess a caregiver's individual household socioeconomic status, local market food prices, and seasonal food availability to recommend practical, affordable, and nutrient-dense meal formulations using accessible local ingredients such as enriching staple porridges with groundnut paste or *mukene* powder).

Furthermore, community health workers face significant operational challenges, including lack of demonstration materials, transport, and supervision, which restricts routine home visits and follow-up on children identified as faltering in growth. One of the key informants stated that:

"We teach mothers about the three food groups during clinic mornings, but the advice is often theoretical. Without practical cooking demonstrations using low-cost, locally available foods, mothers struggle to translate lessons into daily practice." KII 02, Senior Nursing Officer/Health Facility In-Charge.

4.6.2 Socio-Economic Barriers to …..?

Healthcare workers identified household economic vulnerability as the single most critical factor limiting caregivers' ability to achieve Minimum Dietary Diversity and Minimum Acceptable Diet. High market costs for animal-source foods and fresh fruits make them inaccessible to the majority of low-income households. More so, in severe food insecurity, families prioritize bulk staple foods like maize, cassava, or matooke to satisfy hunger, sacrificing meal frequency and micronutrient density for young children. One of the informants cited that:

"Caregivers often understand what a child needs, but poverty limits them. When a mother has 2,000 shillings, she will buy maize flour to feed the entire household rather than an egg or milk specifically for the infant." KII 04, Clinical Officer.

4.6.3 Cultural Beliefs, Maternal Workload, and Child Care

Healthcare Workers highlighted broader socio-cultural norms and time constraints that directly impact responsive complementary feeding practices. Mothers engaged in long hours of agricultural labor or informal trade are often forced to leave infants under the care of young siblings or elderly relatives. Healthcare Workers noted that this results in delayed feeding times, poor meal frequency, and hygienic risks from food kept at room temperature. Certain persistent misconceptions, such as concerns that feeding eggs to young children causes delayed speech or behavioral issues, or that light porridge is safer for infant digestion, continue to undermine nutrient intake. One of the informants cited that:

"Maternal time is a major issue. Mothers go to the field at sunrise and leave porridge in a flask for elder siblings to feed the baby. The feeding frequency drops, and responsive feeding is completely missed." KII 01, Community Health Worker.

CHAPTER FIVE DISCUSSION

5.1 Nutritional Knowledge Regarding Complementary Feeding among Caregivers

Overall, most of caregivers possessed high knowledge regarding the timely initiation of complementary feeding at 6 months of age. Furthermore, caregivers demonstrated strong awareness regarding basic hygiene practices and the necessity of animal-source foods for child growth. However, critical knowledge gaps were identified regarding dietary energy enrichment and food safety handling. Less than half of the caregivers knew that plain cereal porridges should be enriched with energy- and nutrient-dense additives such as peanut paste, milk, or fortified oil. Key Informant Interviews reinforced these observations, noting that while routine antenatal and postnatal health education effectively communicates basic timing and hygiene messages, practical guidance on dietary enrichment and safe food storage remains minimal.

This finding aligns with a similar study in Central Uganda (Kabahenda et al., 2024) which evaluated maternal nutrition knowledge in peri-urban Uganda and found high awareness of complementary feeding timing, but widespread misconceptions regarding energy density, unenriched cereal porridges. Additionally, this finding agreed with a study conducted in Southwestern Uganda, Isingiro District (Mbaziira et al., 2021) that assessed caregiver knowledge among rural mothers, reporting high awareness of animal-source protein importance but low knowledge regarding food safety and storage practices.

However, this study does not agree with studies conducted in Ethiopia for example Asemahagn (2024) evaluated maternal complementary feeding knowledge in Ethiopia, reporting that only 54.5% of mothers knew the correct timing for complementary food initiation, while porridge enrichment knowledge was extremely low (28.0%). The higher knowledge of timely initiation in Komamboga Health Centre III indicates active VHT network and continuous community-level maternal and child health campaigns. Nevertheless, the shared deficit in porridge enrichment knowledge highlights a regional gap in practical complementary feeding education across East Africa.

High general awareness coupled with specific technical knowledge deficits indicates that broad nutrition messaging has reached its limit. Health facility counseling must transition from theoretical lectures to practical, home-based culinary demonstrations. Health education sessions at primary health centers should specifically focus on low-cost food fortification, such

as adding groundnut paste, local seed powders, or oil to cereal gruels, and safe food storage practices to prevent bacterial contamination and energy-density deficits.

5.2 Complementary Feeding Practices among Caregivers of Children aged 6-23 months

Evaluation of WHO IYCF core indicators revealed severe structural deficits in complementary dietary quality and adequacy within the catchment area of Komamboga Health Centre III. Caregiver compliance was characterized by a pronounced disparity between meal quantity and meal quality: while a majority of children achieved the recommended Minimum Meal Frequency, only a minority attained Minimum Dietary Diversity, resulting in overall low achievement of a Minimum Acceptable Diet.

Analysis of individual food groups highlighted an overwhelming reliance on starchy staples, predominantly refined maize, millet, cassava, and sweet potatoes, paired with remarkably low intake of bioavailable animal-source proteins such as flesh foods and eggs. Qualitative insights from healthcare workers corroborated these quantitative trends, identifying household food insecurity, rising food inflation in urban informal markets, and maternal time poverty as the primary structural drivers constraining complementary diet quality.

These findings agree with Bukenya et al. (2019) that evaluated infant feeding practices in Kawempe Division, Kampala, reporting an MDD prevalence of 38.4% and MAD of 29.1%, with starchy staples dominating 91.5% of infant diets. Furthermore, the findings align with the National data from the UDHS (2022) that indicated an increase in national MAD achievement to 27.0%.

However, these findings disagree with Agede et al. (2021) that assessed complementary feeding adequacy in Southern Ethiopia, finding that 21.4% achieved MDD, 53.2% met MMF, and only 12.6% attained MAD, with egg consumption at 12.1%. The higher overall MAD rate in this study is driven by better urban market access and higher continued breastfeeding rates in Kampala peri-urban settings compared to rural Ethiopian pastoralist/agricultural communities. However, the shared deficit in egg and meat consumption highlights a broad Sub-Saharan trend where high market costs render animal proteins inaccessible for complementary feeding.

The gap between Minimum Meal Frequency and Minimum Dietary Diversity proves that infant hunger is being addressed through caloric intake, but micronutrient starvation remains

widespread. Feeding children frequent bowls of unenriched maize porridge satisfies immediate satiety but leads directly to hidden hunger, micronutrient deficiencies, and chronic growth failure. Public health interventions must prioritize market-based micronutrient fortification, promotion of low-cost local protein sources such as mukene/silverfish powder and groundnut paste, and social protection programs targeting vulnerable peri-urban caregivers.

5.3 The nutritional status of children aged 6-23 months

Anthropometric assessment using WHO (2006) child growth standards revealed a substantial burden of undernutrition among infants and young children aged 6-23 months attending Komamboga Health Centre III. Stunting was the predominant form of malnutrition, affecting almost a quarter of the study population. This finding aligns with the national high chronic burden of stunting (24.1%) (UDHS, 2022). Furthermore, this finding agrees with Peri-urban Kampala Studies (Kikafunda et al., 2024; Nabugoomumu et al., 2021).

However, these findings partially agree with Worku et al. (2018) that assessed growth faltering among children aged 6-23 months in Southern Ethiopia, reporting a stunting prevalence of 31.2%, underweight at 18.4%, and wasting at 9.1%. The possible reason for this disparity could be rural agrarian communities in Ethiopia experience heightened seasonal food shortages and lower access to healthcare infrastructure compared to peri-urban Kampala. The relatively lower stunting in our study population is attributed to proximity to urban markets and health facilities, though acute growth faltering remains comparably severe across both settings.

Male infants consistently demonstrated higher rates of stunting, underweight, and wasting than female infants. This agrees with systematic meta-analyses across Sub-Saharan Africa (Wamani et al., 2017; Akombi et al., 2017). Epidemiological literature establishes that male infants under two years of age possess greater biological fragility, higher susceptibility to infectious diseases, and slower immuno-endocrine maturation during early neurodevelopment and growth windows (Agede et al., 2021)

A stunting rate of 24.1% during the critical 6-23-month window indicates that nearly one in every four children at the facility is undergoing irreversible physical and cognitive growth faltering. Because linear growth deficits established before age two are difficult to reverse in later childhood, routine growth monitoring at primary health centers must move beyond passive weight measurements to include routine length/height tracking, early identification of faltering growth, and immediate referral for therapeutic nutritional support.

5.4 Relationship Between IYCF Practices and Child Nutritional Status

Bivariate Chi-Square tests revealed statistically significant associations between key IYCF indicators and child nutritional status ($p < 0.05$). Children who met the Minimum Dietary Diversity demonstrated a significantly lower prevalence of stunting and underweight compared to those who failed to achieve adequate dietary diversity. Similarly, achieving a Minimum Acceptable Diet was significantly associated with lower rates of stunting.

These findings align with national findings from the Uganda Demographic and Health Survey (UDHS 2016 & 2022) and peri-urban studies in Kampala (Bukenya et al., 2019). Furthermore, the findings agree with findings by Marriott et al. (2012) across Sub-Saharan African DHS datasets and regional IYCF analyses. In addition, the finding that consumption of flesh foods or eggs was significantly associated with lower rates of stunting and underweight aligns with empirical literature across East Africa (Headey et al., 2018; Iannotti et al., 2014).

These results highlight that dietary diversity and animal protein intake, rather than meal frequency alone, are key factors associated with healthy child growth. Public health education at primary health facilities must emphasize improving diet quality through low-cost protein additives rather than focusing solely on feeding frequency

CHAPTER SIX: CONCLUSION AND RECOMMENDATIONS

6.1 Conclusion

While basic caloric intake is widely provided, overall complementary feeding adequacy remains severely compromised. Caregiver adherence to Minimum Meal Frequency indicates that basic infant energy demands are largely met in this setting. However, acute deficits in Minimum Dietary Diversity critically restrict the achievement of a Minimum Acceptable Diet. Infant diets are heavily skewed toward starchy staples (maize, cassava, and sweet potatoes), with alarming deficits in micronutrient-dense animal-source foods, particularly flesh foods and eggs. Qualitative evidence confirms that high market prices for animal proteins, household food insecurity, and maternal time poverty are the principal structural drivers enforcing this poor dietary quality.

Undernutrition remains a critical public health challenge among infants attending Komamboga Health Centre III. Stunting constitutes the predominant form of chronic growth failure, exceeding the WHO threshold for high public health concern. This chronic deficit is compounded by notable burdens of underweight and wasting, demonstrating a dual burden of both acute and chronic growth faltering during the first 1,000 days of life. Furthermore, biological male infants consistently exhibit higher vulnerability to growth deficits across all anthropometric indices (HAZ, WAZ, and WHZ) compared to female infants.

Bivariate analysis confirms that dietary quality and animal-protein consumption, rather than meal frequency alone, are the primary determinants of infant growth. Achieving Minimum Dietary Diversity, Minimum Acceptable Diet, and incorporating animal-source foods are significantly associated with a lower prevalence of stunting and underweight ($p < 0.05$). Conversely, meeting Minimum Meal Frequency alone helps maintain short-term weight (wasting) but fails to protect infants against chronic linear growth failure (stunting).

6.2 Recommendations

6.2.1 Recommendations to Caregivers and Community Members

i. Caregivers should actively enrich daily starchy porridges by incorporating affordable, nutrient-dense local ingredients, specifically roasted groundnut paste, and finely ground mukene (silverfish) powder, to increase dietary diversity without incurring prohibitive costs.

ii. Caregivers should adopt small-scale urban agriculture methods, such as vertical sack gardening for green leafy vegetables and micro-scale poultry keeping, to directly improve household access to micronutrient-rich food groups and eggs.

6.2.2 Recommendations to the Healthcare Practitioners and Health Facility Management

i. Healthcare practitioners at Komamboga Health Centre III should integrate rapid 24-hour dietary diversity screening checklists into routine triage at Immunization (EPI) and Growth Monitoring clinics to immediately flag non-compliant caregivers for targeted, one-on-one counseling prior to discharge.

ii. Facility management should reorganize Maternal and Child Health (MCH) clinic operations to include live, practical cooking demonstrations using low-cost local foods, transitioning facility nutrition education from brief, generic lectures into practical skills training on how to prepare enriched complementary meals.

6.2.3 Recommendations to Policy Makers and Government Entities

i. The Ministry of Health (MoH) should revise maternal nutrition education guidelines within the Maternal, Newborn, and Child Health (MNCH) policy framework to explicitly emphasize dietary quality and animal-protein intake alongside caloric meal frequency, directly addressing the widespread practice of feeding high-bulk starchy foods.

ii. Kampala Capital City Authority (KCCA) health department should resource Village Health Teams (VHTs) with standardized IYCF counseling cards and conduct home visits targeting vulnerable households, particularly those caring for male infants who exhibited higher susceptibility to growth deficits.

6.2.4 Recommendations to Future Researchers

i. Future researchers should investigate the nutritional efficacy, bio-availability, microbial safety, and shelf-stability of community-formulated composite flours compared to commercial infant food products in resource-limited Ugandan settings.

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ANNEXES

Annex 1: Consent form

TO PARTICIPATE IN A RESEARCH STUDY TITLED CAREGIVER'S NUTRITION KNOWLEDGE, FEEDING PRACTICES AND NUTRITIONAL STATUS OF CHILDREN 6-23 MONTHS AT KOMAMBOGA HEALTH CENTRE III, KAWEMPE DIVISION, KAMPALA

Dear respondent, I am FLORENCE, a student of Kyambogo University, Bachelor of Sciences in Human Nutrition and Dietetics. I am carrying out a study to assess the relationship between caregiver knowledge, complementary feeding practices, and the nutritional status of children aged 6-23 months at Komamboga Health Centre III, Kawempe Division, Kampala.

I am kindly requesting your cooperation to provide me with the necessary information that will help me accomplish my research study about the above topic. The information got or gathered will strictly be used for academic purposes, that is, for the reward of my Bachelor of Science in Human Nutrition and Dietetics.

In addition, participation in this study is voluntary, and you may choose to withdraw from the study at any point during the interview.

Your assistance will be highly appreciated.

In case of any further inquiries about the study, please contact the research study supervisor, Mangusho Gilbert (Ph.D.), or 0702229915

Participant	**Student researcher**
Signature……	**Signature…….**

Date……../……./2026

Note: The information collected and used for purposes of this study shall be treated with utmost confidentiality by the student researcher and will not be disclosed to any other person outside the purposes of this study. No names except only initials if relevant anywhere on this form.

Annex 2: Data collection tool (Questionnaire)

TO PARTICIPATE IN A RESEARCH STUDY TITLED CAREGIVER'S NUTRITION KNOWLEDGE, FEEDING PRACTICES AND NUTRITIONAL STATUS OF CHILDREN 6-23 MONTHS AT KOMAMBOGA HEALTH CENTRE III, KAWEMPE DIVISION, KAMPALA

Date of Interview: ..... Questionnaire ID: ….....

Name of respondent……...

SECTION ONE: SOCIO-DEMOGRAPHIC INFORMATION

1.1 Age of the respondent (Years): .....

1.2 What is your relationship to the child?

1=Mother

2=Father

3=Grandmother

4=Other: .....

1.2 What is your highest education level attained?

1=None

2=Primary

3=Secondary

4=Tertiary

1.3 What is your marital status?

Married/Cohabiting

Single

Widowed

Divorced/Separated

1.4 What is your primary occupation?

1=Casual work

2=Market vendor

3=Salary income

4=Unemployed

5=Other: .....

1.5 What is your primary source of income?

1= Sale of goods / Business profits

2= Wages / Salary

3= Casual work payouts

4= Support from spouse / Partner

5 = Remittances (Money sent by relatives)

6 = No regular income source

7 = Other (Specify): .....

1.6 What is the total number of people living in your household?

.....

1.7 What is the total number of children 6-23 months in your household

.....

SECTION TWO: NUTRITION KNOWLEDGE ASSESSMENT

2.1 According to health guidelines, at what age should a child start receiving foods in addition to breast milk?

A) 4 months

B) 6 months

C) 9 months

D) 12 months

2.2 What is the recommended duration for exclusive breastfeeding (giving only breast milk and no other food or water)?

A) The first 3 months

B) The first 4 months

C) The first 6 months

D) The first year

2.3 For a child aged 6–8 months who is still breastfeeding, how many times a day should they be fed solid or semi-solid foods?

A) Once a day

B) 2–3 times a day

C) 5 times a day

D) Only when they cry

2.4 For a child aged 9–23 months who is still breastfeeding, what is the minimum number of times they should be fed solid/soft foods and snacks?

A) 1–2 times

B) 3–4 times

C) 6 times

D) 10 times

2.5 What is the correct consistency (thickness) for a child's first porridge at 6 months?

A) Very light and watery so it can be used in a bottle

B) Thick enough to stay on a spoon without running off easily

C) Just plain water with sugar

D) Only clear soup

2.6 To ensure a "diverse diet" for growth, a child should ideally eat from how many different food groups every day?

A) Only 1 group (e.g., just staples)

B) At least 3 groups

C) At least 5 groups

D) All food in the market

2.7 Which of the following is considered an "Iron-Rich" food group essential for preventing anemia in children?

A) Cassava and Sweet potatoes

B) Flesh foods (e.g., Liver, Beef, Fish, or Poultry)

C) Cabbage and Onions

D) Plain maize porridge

*2.*8 Why is it important to include Vitamin A-rich fruits and vegetables (like pumpkins or orange-fleshed sweet potatoes) in a child's diet?

A) To make the food look colorful

B) To improve the child's vision and immune system

C) To make the child sleep better

D) It is not important

*2.*9 When a child starts eating solid foods at 6 months, what should happen to breastfeeding?

A) Breastfeeding should stop immediately

B) Breastfeeding should continue up to 2 years or beyond

C) Breastfeeding should only happen at night

D) The child should only drink cow's milk

*2.*10 What is a common sign that a child may be suffering from "wasting" (acute malnutrition)?

A) The child is very tall for their age

B) The child is very thin for their height

C) The child has many teeth

D) The child is very active

SECTION THREE: INFANT AND YOUNG CHILD FEEDING PRACTICES

	Child 1 (6-23 months)
What is the child's first name?	
3.2 Sex of the child? *Please circle*	1=Male 2=Female

Now I would like to discuss with you about the feeding of children 6-23 months in your household			
3.3	Was [Name of child] ever breastfed? If "no", END questions about retrospective feeding	1=Yes 2=No 3=Don't know	If "No" skip to AA04
3.4	How long after birth was [NAME] first put to the breast? If less than one hour, record "00" hours If less than 24 hours, record hours Otherwise, record days	Immediately Hours (……) Days (…..)	
3.5	In the first two days after delivery, was [NAME] given anything other than breast milk to eat or drink – anything at all like water, infant formula	Yes No	
3.6	Was [NAME] breastfed yesterday during the day or at night?	Yes No Don't know	
3.7	Did [NAME] drink anything from a bottle with a nipple yesterday during the day or at night?	Yes No Don't know	

Now I would like to ask you about food that [NAME] had yesterday during the day or at night. Please tell me about all drinks and food, whether [NAME] had them at home, or somewhere else.

Item	Time	Meal	Description of item before cooking	Amount of food eaten		Source
			Description of ingredients	Amount eaten		
				Household measures	Kg/liters	

When the respondent recall is complete, fill in the food groups based on the information recorded above. For any food groups not mentioned, ask the respondent if a food item from this group was consumed.

S/No	Food categories	Examples	YES=1 NO=0
A	**Foods made from grains**	corn/maize, rice, wheat, sorghum, millet or any other grains or foods made from these (e.g. bread, noodles, porridge or other grain products) + *insert local foods e.g. ugali, nshima, porridge or paste*	
B	**White roots and tubers and plantains**	*White potatoes, white yams, manioc/cassava/yucca, cocoyam, taro or any other foods made from white-fleshed roots or tubers, or plantains*	
C	**Pulses (beans, peas and lentils)**	*Mature beans or peas (fresh or dried seed), lentils or bean/pea products, including hummus, tofu and tempeh*	
D	**Nuts and seeds**	*Any tree nut (almond, cashew), groundnut/peanut or certain seeds, or nut/seed "butters" or pastes, pumpkin seeds, wild mango, hibiscus seeds, flaxseeds, shea butter seeds, sesame*	
E	**Milk and milk products**	*Milk, cheese, yoghurt or other milk products but NOT including butter, ice cream, cream or sour cream*	
F	**Organ meat**	*Liver, kidney, heart or other organ meats or blood-based foods, including from wild game*	
G	**Meat and poultry**	*Beef, pork, lamb, goat, rabbit, wild game meat, chicken, duck or other bird*	
H	**Fish and seafood**	*Fresh or dried fish, shellfish or seafood*	
I	**Eggs**	eggs from chicken, duck, guinea fowl or any other egg	
J	**Dark green leafy vegetables**	*List examples of any medium-to-dark green leafy vegetables, including wild/foraged leaves, carrot greens, cassava greens, broccoli, spinach, dodo, pumpkin greens, sweet potato greens, okra, carrot greens*	

K	**Vitamin A-rich vegetables, roots and tubers**	*Pumpkin, carrots, squash or sweet potatoes that are yellow or orange inside,*	
L	**Vitamin A-rich fruits**	*Ripe mango, ripe papaya, apricot, passion fruits, peaches, musk melon, Locust bean fruit/pulp*	
M	**Other vegetables**	Beans (various) when eaten as fresh pods, cabbage, bitter melons, Cauliflower, celery, cucumber, egg plants, lettuce, peas, okra, nakati, onions, mushrooms, snake gourd, tomato	
N	**Other fruits**	Apple, banana, black current, blackberry, blue berry, cactus pears, figs, grapes fruits, grapes, guava, jackfruit, lemon, lime	

SECTION FOUR: ANTHROPOMETRIC MEASUREMENT OF THE CHILD

Interviewer: Ensure equipment is calibrated before measurement.

What is the child's age……..

CATEGORY	**Reading 1**	**Reading 2**	**Average reading**
Weight (Record to nearest 0.1 kg)			
Height (Record to nearest 0.1 cm)			
MUAC (Record to nearest 0.1 cm)			

MUAC Color Category:

1=Green (Normal)

2=Yellow (MAM)

3=Red (SAM)

Bilateral Pitting Edema:

1=Present

2=Absent

THANKS FOR YOUR PARTICIPATION

Annex 3: Key Informant Interview Guide

TO PARTICIPATE IN A RESEARCH STUDY TITLED CAREGIVER'S NUTRITION KNOWLEDGE, FEEDING PRACTICES AND NUTRITIONAL STATUS OF CHILDREN 6-23 MONTHS AT KOMAMBOGA HEALTH CENTRE III, KAWEMPE DIVISION, KAMPALA

Date of Interview.....

SECTION ONE: PARTICIPANT PROFILE

Participant Code: .....

Professional Role: .....

Years of Experience at Komamboga HC III: .....

SECTION TWO: INTERVIEW QUESTIONS

Current Nutritional Landscape

1. In your experience at this facility, what are the most common nutritional deficiencies or conditions you observe among children aged 6–23 months?

2. How would you describe the trend of malnutrition cases (stunting or wasting) in the Komamboga catchment area over the last year?

Caregiver Knowledge and Practice

3. Most caregivers receive nutritional counseling during ANC and YCC. In your opinion, how much of this information is actually put into practice at home?

4. What specific aspects of complementary feeding (e.g., timing, diversity, or frequency) do caregivers find most difficult to follow? Why?

The Urban Context & Maternal Employment

5. Komamboga serves many informal settlements (slums). How do you think the living conditions in these areas impact a child's nutritional status?

6. Many mothers in Kawempe are market vendors or casual laborers. How does their work schedule affect how their children are fed?

7. Who usually takes over feeding when the primary mother is away, and how does this affect the quality of the child's diet?

Cultural and Socio-Behavioral Drivers

8. Are there specific cultural beliefs, taboos, or myths in this community that prevent caregivers from giving certain nutrient-dense foods (e.g., eggs, fish, or certain vegetables)?

9. What role do fathers or other household members play in the decision-making process regarding infant feeding?

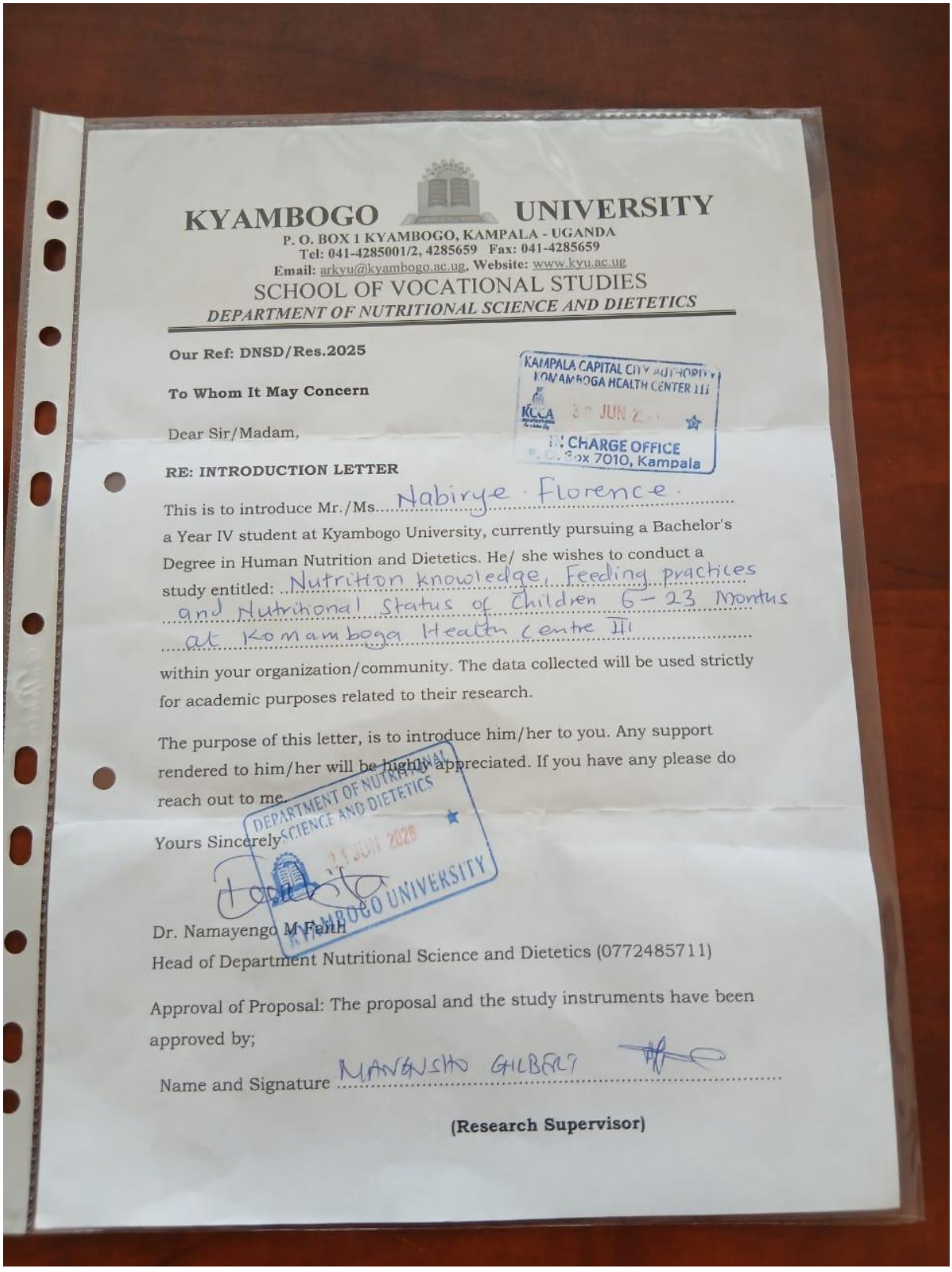
Facility and Systemic Barriers

10. What are the main challenges you face as a healthcare provider in ensuring that nutritional education effectively reaches the caregivers?

11. What resources or interventions (that are currently missing) do you think would most improve the nutritional outcomes of the children attending this clinic?

THANKS FOR YOUR PARTICIPATION

Annex 4: Research Ethical Clearance Letter



KYAMBOGO UNIVERSITY

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SCHOOL OF VOCATIONAL STUDIES

DEPARTMENT OF NUTRITIONAL SCIENCE AND DIETETICS

Our Ref: DNSD/Res.2025

KAMPALA CAPITAL CITY AUTHORITY
KOMAMBOGA HEALTH CENTER III
KCCA
30 JUN 2025
IN CHARGE OFFICE
P.O. Box 7010, Kampala

To Whom It May Concern

Dear Sir/Madam,

RE: INTRODUCTION LETTER

This is to introduce Mr./Ms. Nabirye Florence a Year IV student at Kyambogo University, currently pursuing a Bachelor's Degree in Human Nutrition and Dietetics. He/ she wishes to conduct a study entitled: Nutrition knowledge, Feeding practices and Nutritional Status of Children 6-23 Months at Komamboga Health Centre III within your organization/community. The data collected will be used strictly for academic purposes related to their research.

The purpose of this letter, is to introduce him/her to you. Any support rendered to him/her will be highly appreciated. If you have any please do reach out to me.

DEPARTMENT OF NUTRITIONAL SCIENCE AND DIETETICS
25 JUN 2025
KYAMBOGO UNIVERSITY

Yours Sincerely

Dr. Namayengo M Faith
Head of Department Nutritional Science and Dietetics (0772485711)

Approval of Proposal: The proposal and the study instruments have been approved by;

Name and Signature MANGUSHO GILBERT

(Research Supervisor)